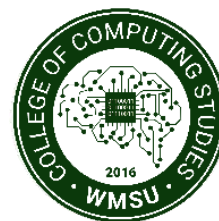




Republic of the Philippines
Western Mindanao State University
COLLEGE OF COMPUTING STUDIES
DEPARTMENT OF INFORMATION TECHNOLOGY



SAFENEST: CONTACTLESS BASED REAL-TIME FLOOD DETECTION AND MONITORING SYSTEM

A CAPSTONE PROJECT

Presented to the Faculty of the

College of Computing Studies
Western Mindanao State University

In Partial Fulfillment
of the Requirements for the Degree
Bachelor of Science in Information Technology

By

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CERTIFICATE OF PANEL APPROVAL

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EXECUTIVE SUMMARY

The capstone project titled "SafeNest: Contactless Based Real-time Flood Detection and Monitoring System" focuses on developing an innovative solution to address the challenges of monitoring water levels in flood-prone or resource-sensitive areas. The project aims to enhance community preparedness and safety by integrating modern IoT technologies, data analytics, and user-friendly interfaces.

The system employs an HC-SR04 ultrasonic sensor to measure water levels, with the data displayed on an LCD1602 screen in percentage format and categorized with LED indicators: Green (Low), Yellow (Medium), and Red (High). The data is transmitted to Firebase in real time, allowing for continuous storage and retrieval. A responsive dashboard, developed using JavaScript, presents the information in graphical form for use by administrators and community leaders such as Kagawad or SK Kagawad.

The project methodology combines hardware assembly, programming, and data visualization techniques. The outcomes include accurate water level readings, seamless data transfer to Firebase, and a user-friendly dashboard. This system has the potential to improve disaster risk reduction efforts and empower local leaders with actionable insights for better resource management and community safety.

DEDICATION

The capstone project "SafeNest: Contactless Based Real-time Flood Detection and Monitoring System" is dedicated to researcher's advisor, Mr. Whesley C. Timpangco, researcher's parents, whose unwavering support and encouragement have been the foundation throughout the journey. To the mentors and professors, who have guided the researchers with their wisdom and expertise, they owe their deepest gratitude. Lastly, they dedicate the work to the communities they aim to serve, with the hope that the project will contribute to a safer and more resilient future.

ACKNOWLEDGEMENT

The researchers would like to express their heartfelt gratitude to all those who have supported them in the success and completion throughout of their capstone project:

- First and foremost, to the Almighty God, who gave them the determination and willpower to start and successfully finish their research study;
- To the researchers' ever supportive parents, who have been there ready to aid the students when they needed help;
- To the researchers' Thesis Adviser, Mr. Ceed Lorenzo for helping and guiding them all throughout the research study;
- To researchers' Class adviser, Mr. Whesley C. Timpangco for supporting and motivating them for the entire duration of their research study.

The project would not have been possible without the collective efforts and encouragement of everyone involved. Thank you for being a part of the journey.

TABLE OF CONTENTS

TITLE	i
APPROVAL SHEET	ii
EXECUTIVE SUMMARY	iii
DEDICATION	iv
ACKNOWLEDGEMENT	v
LIST OF TABLES	xi
LIST OF FIGURES	xii
LIST OF APPENDICES	xiii
LIST OF ABBREVIATIONS	xiv

Chapter	Page
1 INTRODUCTION	9
1.1 Project Context	9
1.2 Statement of the Problem	10
1.3 Purpose and Description	10
1.4 Objectives	10
1.4.1 Specific Objectives	Error! Bookmark not defined.
1.5 Scope and Limitation	11
1.6 Significance of Study	11
1.7 Definition of Terms	8
2 REVIEW OF RELATED LITERATURE AND SYSTEMS	9
2.1 Introduction	9
2.1.1 RRL 1	Error! Bookmark not defined.
2.1.2 RRL 2	Error! Bookmark not defined.
2.1.3 RRL 3	Error! Bookmark not defined.
2.1.4 RRL 4	Error! Bookmark not defined.
2.2 Summary	Error! Bookmark not defined.
2.3 Synthesis	Error! Bookmark not defined.
3 TECHNICAL BACKGROUND	10
3.1 Conceptual Framework	10
3.2 Software and Hardware Requirements	12
4 DESIGN AND METHODOLOGY	14
4.1 Introduction	14

4.2	Requirements Analysis.....	Error! Bookmark not defined.
4.2.1	Context Diagram	17
4.2.2	Data Flow Diagram	Error! Bookmark not defined.
4.2.3	Flowchart.....	Error! Bookmark not defined.
4.2.4	Use Case	Error! Bookmark not defined.
4.3	Requirement Specification	23
4.3.1	Functional Requirements.....	Error! Bookmark not defined.
4.3.2	Non-Functional Requirements	Error! Bookmark not defined.
4.4	System Design.....	Error! Bookmark not defined.
4.4.1	Database Design	Error! Bookmark not defined.
4.4.2	Entity Relationship Diagram	Error! Bookmark not defined.
4.4.3	Database Fields.....	Error! Bookmark not defined.
4.4.4	User Interface Design.....	Error! Bookmark not defined.
5	DEVELOPMENT AND TESTING	44
5.1	Development	44
5.2	Integration of Technology	44
5.3	Testing.....	45
5.3.1	Alpha Testing	48
5.3.2	Beta Testing.....	2
5.3.3	User Acceptance Testing.....	7
5.4	Implementation Plan.....	8
5.4.1	Infrastructure/Deployment	8
5.4.2	Processes/Policies/Personnel.....	8
6	RESULT AND DISCUSSIONS.....	10
6.1	Introduction	10
6.2	Summary of Findings	10
7	CONCLUSION AND RECOMMENDATIONS	15
7.1	Conclusions	15
7.2	Recommendations	15
	REFERENCES	17
	APPENDICES	Error! Bookmark not defined.
	CURRICULUM VITAE	Error! Bookmark not defined.

LIST OF TABLES

Table	Page
2.3 Synthesis	11
6.1 Number of Respondents	36

LIST OF FIGURES

Figure	Page
3.1 Conceptual Framework	14
4.1 Software Development Life Cycle	17
4.2 Flowchart	18
4.3 Use Case	19
4.4 Context Diagram	23
4.5 Data Flow Diagram	23

LIST OF APPENDIXES

Appendix	Page
Appendix A: Data Gathering	53
Appendix B: Source Code	54
Appendix C: Testing	55
Appendix D: Evaluation Form	57
Appendix E: Gantt Chart	60
Appendix F: Use Case	61

CHAPTER 1

INTRODUCTION

1.1 Project Context

Tumaga, a barangay in Zamboanga City, Philippines, is frequently affected by flooding due to its geographical location and climatic conditions, particularly for homes situated near the riverside. In recent years, severe flooding has resulted in significant casualties in the area. Most notably, in June of this year, the floodwaters once again reached critical levels, leading to widespread damage to properties and infrastructure, and tragically, the loss of lives.

During periods of flooding, The Tumaga River, flowing through the river often overflows its banks, inundating nearby homes and infrastructure, especially those situated close to the riverside. These floods have led to substantial severe flooding, property damage, disruption of daily life, and, in severe cases, loss of lives. The river's swelling exacerbates the vulnerability of the community. According to PAGASA, when yellow rainfall warning is issued, the intensity is between 7.5 mm and 15 mm per hour which flooding is possible in low-lying areas and communities are advised to be aware of potential flooding. When orange notification alerts, the intensity is between 15 mm and 30 mm per hour where at this level, there is a high likelihood of flooding and rivers such as the Tumaga River may approach critical levels, indicating an increased risk of overflow. The most critical level is the red warning alert where when it exceeds 30 mm per hour; widespread flooding is expected, evacuation is strongly advised where its likely to overflow significantly, threatening lives and property.

Despite awareness of the potential for river overflow, residents living near the riverbanks, as well as the National Disaster Risk Reduction and Management Council (NDRRMC), often face challenges in determining precisely when the overflow begins. This delay in real-time information poses a significant risk, as it hinders the timely implementation of evacuation and safety measures. Although the community is informed of the likelihood of flooding, the lack of immediate, real-time updates regarding the river's status leaves them vulnerable to the sudden onset of dangerous conditions. The absence of timely alerts

intensifies the risk, making it difficult to react swiftly and prevent harm during critical flood events.

SafeNest aims to address these critical challenges by developing a comprehensive flood detection and response system tailored specifically for barangay Tumaga. The primary goals of SafeNest are to enhance early warning capabilities and improve public awareness and preparedness; streamlines emergency response efforts.

1.2 Statement of the Problem

The recurring flooding in Barangay Tumaga, Zamboanga City, poses a significant threat to the safety and well-being of Zamboangueños, particularly those living in close proximity to the Tumaga River. Despite the community's awareness of the potential for overflow, there is a lack of real-time manner of information regarding the river's water levels and flood alerts. Furthermore, the absence of a hardware infrastructure to monitor in real-time effective disaster management system exacerbates the challenges faced by local authorities and emergency responders in coordinating timely evacuation and relief efforts.

1.3 Purpose and Description

The purpose of this study is to design and develop a flood detection tailored for Barangay Tumaga. The system aims to provide real-time monitoring and early warning to local authorities, such the barangay officials. The real-time system enables timely and effective responses to potential flood threats.

1.4 Project Objectives

This section presents the general objective and the specific objectives of the study.

a. General Objective

To develop a flood detection system entitled, “SafeNest: Contacless Based Real-time Flood Detection and Monitoring System,” Zamboanga City that will monitor water level of the rivers and generate public alert when it reaches a critical level to a potential flood risk.

b. Specific Objectives

- To design a water level monitoring device using an Ultrasonic sensor.
- Develop an embedded program for ESP32 microcontroller that will reads water level using the ultrasonic sensor that uploads the data to the remote server.
- To create a notification system using SMS to alert barangay officials the potential flooding.
- To identify a potential threshold for water level to that may cause flooding.
- Develop an online web application to display the environmental data and generate alerts when certain thresholds are met.

1.5 Scope and Limitation

The scope of the study will focus on measuring water levels real-time using an advanced monitoring system that incorporates an Ultrasonic Sensor HC-SR04. The system will be deployed along the Tumaga River to continuously collect data on water levels and transmit the information to a centralized dashboard accessible to the local authorities, the barangay officials of Tumaga. The study analyzes the correlation between river water levels and the occurrence of flooding that assess the monitoring system in providing timely alerts and enhancing community awareness about the flood risk.

1.6 Significance of Study

The study titled “SafeNest: Contactless Based Real-time Flood Detection and Monitoring System” carries significant importance for various stakeholders in Barangay Tumaga and beyond, as it aims to develop a system that facilitates the monitoring of environmental data.

- **People.** The implementation of a real-time flood monitoring system will empower residents with timely information about rising water levels, enabling them to take proactive measures to protect their lives and property. Increased awareness and preparedness can significantly reduce the impact of flooding, fostering a greater sense of safety within the community.

- **Barangay Officials.** Provides essential data and tools to enhance disaster management strategies. By utilizing real-time data on water levels, officials can make informed decisions regarding evacuation plans, resource allocation, and emergency response efforts, ultimately improving the overall effectiveness of disaster preparedness initiatives.
- **Residents.** Reducing their vulnerability to flood-related risks. Access to timely alerts and information will enable families to implement safety measures and evacuate, when necessary, thus minimizing the potential for loss of life and property damage.
- **Future Researchers.** Serves as a valuable resource for future researchers exploring flood monitoring systems and disaster management in similar geographical contexts. The methodologies and findings can inform subsequent studies, providing insights into effective practices for real-time data collection, analysis, and community engagement.

1.7 Definition of Terms

Term	Definition
1. Flood Detection System	A system that detects potential flood risks and provides early warning signs.
2. Public Awareness	The level of knowledge and understanding the general public has about a particular issue or set of issues.
3. Preparedness	Measures taken to prepare for and reduce the effects of disasters.
4. Ultrasonic Sensor	Devices are used to measure the depth of water in the rivers by emitting sound waves and convert into distance.
5. Thresholds	Pre-determined water level limits that, when reached or exceeded, trigger alerts for potential flood risks.
6. Early Warning System	A system that provides timely and effective information through identified institutions to individuals and communities at risk of floods, enabling them to prepare and act appropriately

Term	Definition
7. Emergency Response	Actions taken by local authorities and emergency services to manage and mitigate the effects of a flood, including evacuation, rescue, and provision of relief
8. Public Announcements	Information is disseminated to the public through various media, including loudspeakers, radio, television, and social media, to inform and instruct residents during a flood emergency
9. Pilot Testing	A preliminary implementation phase where the flood detection and response system are tested on a small scale to identify any issues and gather feedback for improvements.
10. Casualties	People who are injured, killed, or missing during a disaster.
11. SMS Notification Alerts	Short message service (SMS) alerts are sent via mobile phones to inform barangay officials about flood warnings and emergency instructions.
12. Residents	Refers to the composed communities.
13. Barangay Tumaga	A small administrative division in Tumaga, Zamboanga City, Zamboanga Del Sur, Philippines.
14. Disaster Resilience	The ability of a community to withstand and recover from flooding events through effective preparation, response, and adaptation strategies.
15. Real-time Monitoring	The continuous observation and recording of data, such as water levels, as it occurs, allows for immediate analysis and response.

Table 1: Definition of Terms

CHAPTER 2

REVIEW OF RELATED LITERATURE AND SYSTEMS

2.1 Introduction

The review of related literature and systems focused on enhancing emergency response systems using ESP32 offers a thorough understanding of current emergency response tools, their functions, and the technological progress that has influenced this area. This review pinpoints essential features and methodologies in comparable systems, emphasizing their advantages and drawbacks to guide the development of the present project. By examining previous studies, we gain a clearer picture of the emergency response system landscape, allowing us to recognize potential areas for innovation and enhancement.

The need for efficient, scalable, and cost-effective emergency response systems has never been more apparent in the face of increasing natural disasters and emergencies. The advent of microcontroller platforms like ESP32 has opened new avenues for developing innovative solutions to enhance these systems. ESP32, known for its integrated WI-FI and dual-mode Bluetooth capabilities and for its affordability, versatility, and ease of use, has become popular for prototyping and deploying various emergency response applications

The following literature and systems exhibit comparable structures and thematic relevance.

2.1.1 IoT-Based Flood Detection and Alert System using Wi-Fi and GSM Module

The study presents an IoT-based flood detection and alert system that uses Wi-Fi and GSM modules to detect water levels and send alerts to authorities and residents. The system consists of a sensor node that measures water levels using an ultrasonic sensor, a Wi-Fi module that sends data to a cloud server, and a GSM module that sends SMS alerts to authorities and residents. The system was tested in a real-world scenario and showed accurate detection of water levels and timely alerts.

- ESP32 Micro-controller: as the micro-controller that is low-cost and low power device that can connect to the internet.
- TOF10120 Laser Sensor: to measure water depth levels accurately.

- LED Indicators: to categorize water depth levels into three ranges: alert, warning, and danger.
- Buzzer: to provide an audible alert when the water depth level reaches a certain threshold.
- LCD Display: to show the water depth level readings
- ESP32 Cam: to provide visual monitoring of the flood situation.
- Blynk IoT Application: to monitor and receive notifications on water depth levels remotely.
- Real-Time Monitoring: to provides real-time monitoring of water depth levels, enabling authorities and residents to take prompt action.
- Remote Alerting: it sends notifications to authorities and residents through the Blynk IoT application, enabling them to take action remotely.

“The system acknowledges the relevance of IoT-based flood monitoring and early warning systems on reducing the impact of floods, and builds upon their findings to develop a more effective system.” (*Ahmed et al., 2023*)

2.1.2 IoT-based Flood Monitoring System using ESP32

The IoT-based flood monitoring system developed in this work uses the ESP32 microcontroller, TOF10120 laser sensor, LEDs, an LCD, and a buzzer to detect water depth levels and issue early flood warnings. The system categorizes flood severity into three levels: alert, warning, and danger, corresponding to green, yellow, and red LEDs, respectively, and uses a buzzer to signal danger. An ESP32 cam is integrated for visual monitoring, and the system transmits data wirelessly via Wi-Fi to the Blynk app, which enables real-time remote monitoring and notifications for users and authorities. The system offers early detection, alerts, and remote flood management, helping mitigate flood impacts through timely responses.

- ESP32 Micro-controller: wirelessly transmits data to the Blynk app for real-time water depth monitoring and visualization via a smartphone.
- TOF10120 Laser Sensor: to measure water depth, categorizing it into three severity levels: alert (green LED), warning (yellow LED), and danger (red LED).
- Blynk IoT platform: for remote monitoring. Alerts are sent to authorities and nearby people via the app.

- Buzzer and LED Indicators: to categorize water depth levels into three ranges: alert, warning, and danger.
- ESP32 Cam: to visually monitor water levels in real-time, further enhancing the system's monitoring capabilities.

“The system is highly efficient in mitigating the impact of floods by offering early warnings and enabling proactive measures to reduce damages.” (*Jais Lias et al., 2024*)

2.1.3 Development of a Prototype of River Water Level Monitoring System using ESP32 based on IoT for Flood Mitigation

The system consisted of an ESP32 microcontroller, an ultrasonic sensor, and a Wi-Fi connection to transmit data to the Blynk application and the Thingspeak website. It was able to transmit data wirelessly to the Blynk application and the Thingspeak website, providing real-time information on river water levels.

- ESP32: acts as the microcontroller as the brain of the system, which is a Wi-Fi and Bluetooth-connected Smart on Chip with a simple design.
- Ultrasonic Sensor: to measure the water level in the river.
- IoT Technology: to transmit data wirelessly to the Blynk application and the Thingspeak website.
- Real-time monitoring: to provide real-time information on river water levels, enabling early warning systems and timely evacuation.
- LED Indicators: to provide visual warnings of water level changes.
- Arduino IDE Software: to program the ESP32 microcontroller.

“The river water level monitoring system utilizes an ESP32 microcontroller and ultrasonic sensors to continuously measure water levels in real-time. The system updates flood risk statuses through the Blynk and Thingspeak platforms, ensuring that any fluctuations in water levels promptly trigger updated alerts to users and relevant authorities.” (*Pramono et al., 2024*)

2.1.4 Design Analysis of an IoT Based Early Flood Detection and Alert System

The system aims to develop and implement an Internet of Things (IoT) based early flood detection system that utilizes weather API parameters to trigger SMS alerts. The system employs an ESP32 microcontroller, integrated with sensing circuitry to measure water levels, and a GSM module responsible for sending timely SMS alerts to both residents and emergency response units in the event of critical flooding. Additionally, the system incorporates a liquid crystal display (LCD) to indicate the current system status and provide real-time weather parameter readings from the digital temperature and humidity module 11 (DHT11). To facilitate efficient monitoring and data access, the system is integrated with the Blynk mobile app which offers unrestricted access to real-time water level weather parameter data for both the controller chief and the forecasting unit.

- ESP32 WROOM Microcontroller: central control unit of the system
- Floater and Potentiometer: measures water levels with precision, equipped with two thresholds for alerting purposes.
- DHT11 Sensor: provides real-time temperature and humidity readings.
- Buzzer: alerts operators when water surpasses the first threshold.
- SIM800 GSM Module: responsible for sending timely SMS alerts to both residents and emergency response units in the event of critical flooding.
- 16x2 LCD Display: shows water levels, temperature, and humidity data in real-time.
- Blynk Mobile App: allows remote monitoring of real-time data (water level and weather) from smartphones.
- 7.5V Rechargeable Battery: powers the system with voltage regulation through a buck converter.

“The system uses IoT-based flood detection and alerting mechanisms to monitor real-time data, including water levels and weather conditions. It uses sensors to track changes in parameters like water height and humidity. Once certain thresholds are met, the system updates to reflect those changes by sending SMS alerts to residents and emergency response units, indicating the need for proactive measures to avoid damage or risk.” (Okeke, 2024)

Table 2.0 Review of Related Works and System Comparison Table

Features	IoT-Based Flood Detection and Alert System using Wi-Fi and GSM Module	IoT-based Flood Monitoring System using ESP32	Development of a Prototype of River Water Level Monitoring System using ESP32 based on IoT for Flood Mitigation	Design Analysis of an IoT based Early Flood Detection and Alerting System	SafeNest: Contactless Based Real-time Flood Detection and Monitoring System
User Account Login/Logout	✓				✓
Microcontroller	✓	✓	✓	✓	✓
Ultrasonic Sensor			✓		✓
Data Transmission	✓	✓	✓	✓	✓
Alert Mechanism	✓	✓	✓	✓	✓
Programming Languages	✓	✓	✓	✓	✓
Power Supply			✓	✓	✓
Dashboard	✓	✓	✓	✓	✓
Real-time Monitoring	✓	✓	✓	✓	✓
Scalability	✓	✓	✓	✓	✓
Database					✓

2.2 Summary

The IoT-based Flood Detection and Alert System uses ESP32 microcontroller, an ultrasonic sensor, the TOF10120 laser sensor, a LED indicators, buzzer, and LCD display, and a ESP32 Cam; it detects water levels and sends alerts via Wi-Fi and GSM modules. The system uses the Blynk IoT application for real-time monitoring and sends notifications to authorities and residents via SMS. The features of the system provide a real-time monitoring, remote alerting, visual monitoring (via camera), and multi-level alert indicators. The IoT-based Flood Monitoring System using ESP32 also uses the ESP32 microcontroller, a TOF10120 laser sensor, the LED indicators, a buzzer, and a ESP32 Cam where it also measures water depth levels, categorizes them into three severity levels (alert, warning, danger), and sends real-time data to the Blynk app for remote monitoring and notifications. LED indicators for flood severity, buzzer for alerts, visual monitoring via camera, and Wi-Fi-based data transmission. Another system, the Development of a Prototype of River Water

Level Monitoring System using ESP32 that also has a similar functionality that measures river water levels using an ultrasonic sensor and transmits real-time data to the Blynk Application and Thingspeak website; alerts are issues based on water level changes. Lastly, the Design Analysis of an IoT-based Early Flood Detection and Alerting System uses ESP32 WROOM microcontroller, a floater and potentiometer, an DHT11 sensor, GSM module, LCD display, and a buzzer; measures water levels and weather parameters (temperature and humidity), sends timely SMS alerts to residents and emergency units using the GSM modules.

2.3 Synthesis

The common theme across these studies is the utilization of IoT and microcontroller technologies to create effective flood monitoring and alert systems. Each system employs a combination of sensors to monitor water levels and environmental conditions, processes the data using microcontrollers, and sends timely alerts to residents and authorities through various communication methods such as GSM modules, Android apps, and cloud services. The SafeNest addresses this pressing need by developing a real-time flood monitoring and detection system using ESP32-based; this project integrates IoT devices, cloud-base services, and real-time monitoring to provide early warning alerts to local authorities helping mitigate the impact of flood-related disasters. The system incorporates a combination of hardware such as ESP32 microcontroller, ultrasonic sensors. LED indicators, LCD displays. For software, such as: Firebase for real-time data storage, MySQL for user management, a web-based dashboard for monitoring. This integration allows for continuous monitoring of water levels and timely notifications when thresholds are breached.

CHAPTER 3

TECHNICAL BACKGROUND

3.1 Conceptual Framework

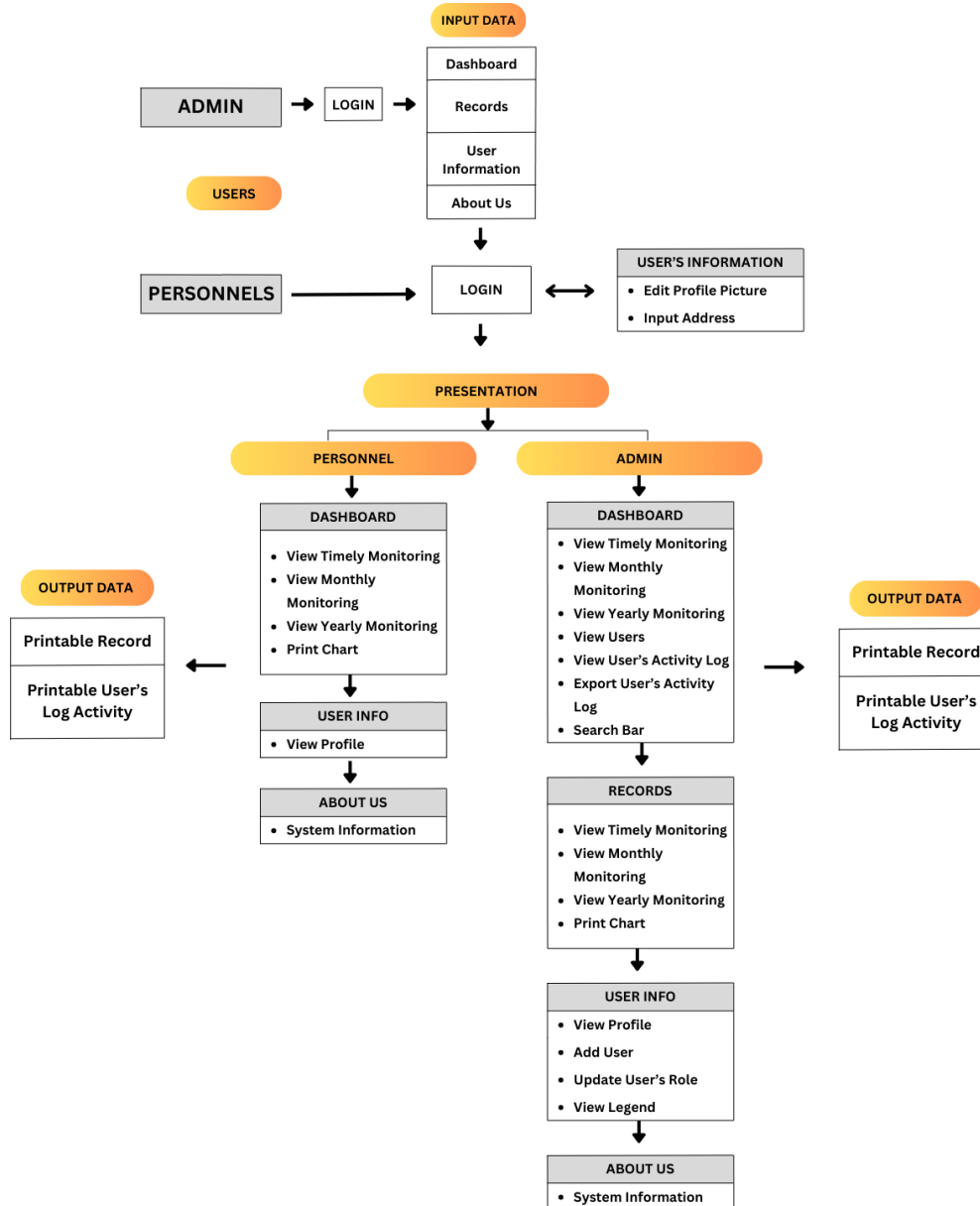


Figure 3.1. Conceptual Framework of the Software System

The Flooding Monitoring in Barangay Tumaga is set into various categories; Admins/Developers, Personnel/Barangay Officials, Kagawad, and SK Kagawad. Personnels are tasked to monitor the water level of the said place. Using the developed system, it becomes more efficient for faster action rather than manual procedure. Data are converted into graphical form and user information will be stored securely in the database.

The system will be in the form of a web-based type. The development process will require various languages such as JavaScript, HTML, CSS, PHP, and for the database, the team will be using MySQL for storing of user's data while Firebase used for real-time data in the dashboard. This will secure all stored information and expect faster responses. The system will be in a modern type of technology which is expected to deliver scalability and accuracy reports.

The framework is divided into three main layers: Monitoring, Presentation, and Results.

- **Monitoring Layer**

The monitoring layer will be the first step which in this part is where you get the data of live monitoring of the water level of Barangay Tumaga. This layer will send the data collected to the presentation part.

Barangay Officials: Required to log in with their credentials on the website, they can view the live monitoring graphs, and generate reports.

Administrators: Administrators have access to manage the overall system, this includes both hardware, as well as website.

- **Presentation Layer**

The presentation layer shows the actual graphs of live monitoring, as well as previous records of flooding. This layer includes database records, sorting, information about us, and live monitoring.

Database Records: The database is responsible for storing all the data from the previous to the present for the system to work. Those previous flooding records are all stored in a database. This indicates the total result of the water level monitoring to ensure secure safety.

Sorting: This function is necessary to make it very simple since the officials are dealing with the graph results. This will sort the years or months of graphs.

Live Monitoring: Data will be collected from the live retrieval and converted into graphical form. Various colors will be set as guides and indicate their level. Day-to-day observation is needed. Once the water level reaches a specific threshold, this will automatically trigger a critical level by sending an SMS notification.

- **Results Layer**

This layer will show the output result from the live monitoring. This includes printable documents such as graphs and previous records.

SMS Alert Warning: An SMS will be sent to the authorized users of Barangay Tumaga if only the water level hits the minimum point.

Printables: Data that is retrieved from the monitoring can be printables.

- **User Interactions**

Barangay Officials: Interact with the website to monitor water level, and alarm requests, generate results, and perform evaluations.

Administrators: Oversee the system operations in both hardware and website, ensure data is stored, confirm alarms, manage graphs, and generate printable results.

3.2 Software and Hardware Requirements

1. Operating Systems

Server: A reliable operating system such as Node.js Server or Windows Server is required to host the web server and MySQL database.

The application should be accessible via modern web browsers. Also, will work on various operating systems such as Windows and macOS.

2. Web Server

XAMMP: A local host or server to test its website and clients via computers and laptops before releasing it to the main server.

3. Database Management System

MySQL: Chosen for its reliability and scalability, MySQL will manage the data storage needs, ensuring efficient data retrieval and manipulation.

4. Development Tools

Arduino Hardware: Arduino IDE, ESP32, and Twilio for developing the flooding hardware system.

Web Development Tools: Visual Studio Code for developing web applications using HTML, CSS, PHP, and JavaScript.

5. Programming Languages and Libraries

C++: Used for Arduino IDE in developing hardware flooding sensors.

HTML, CSS, JavaScript: For building the web application's front end.

XAMMP, Firebase: For the web server and storing of data

PHP and JavaScript: For backend services.

MySQL and Firebase: For database and other user

CHAPTER 4

DESIGN AND METHODOLOGY

This chapter presents the design process and methodology of the development of the project.

4.1 Introduction

The project was guided by the Agile software development methodology, which provided a flexible and iterative framework suitable for the continuous integration and testing required by an IoT-based flood monitoring system. The Agile approach, allowed for incremental progress, regular feedback from testing, and swift adaptations in response to evolving project requirements.

The project integrates modern technologies, including the ESP32 microcontroller, ultrasonic sensor, LED indicators, LCD displays, and a cloud-based dashboard to enable real-time monitoring of flood levels in Barangay Tumaga. These methodologies facilitated the refinement of system components and the seamless integration of hardware and software functionalities through iterative development cycles, known as sprints.

This section outlines the design methodology utilized in the study, detailing the key components that make up the system. Additionally, it elaborates on the flowchart algorithm and presents the block and circuit diagrams to illustrates the system's structure and operation. A top-down approach is adopted to ensure clarity and coherence in the explanation.

4.2 IoT Hardware Development

4.2.1 Block Diagram

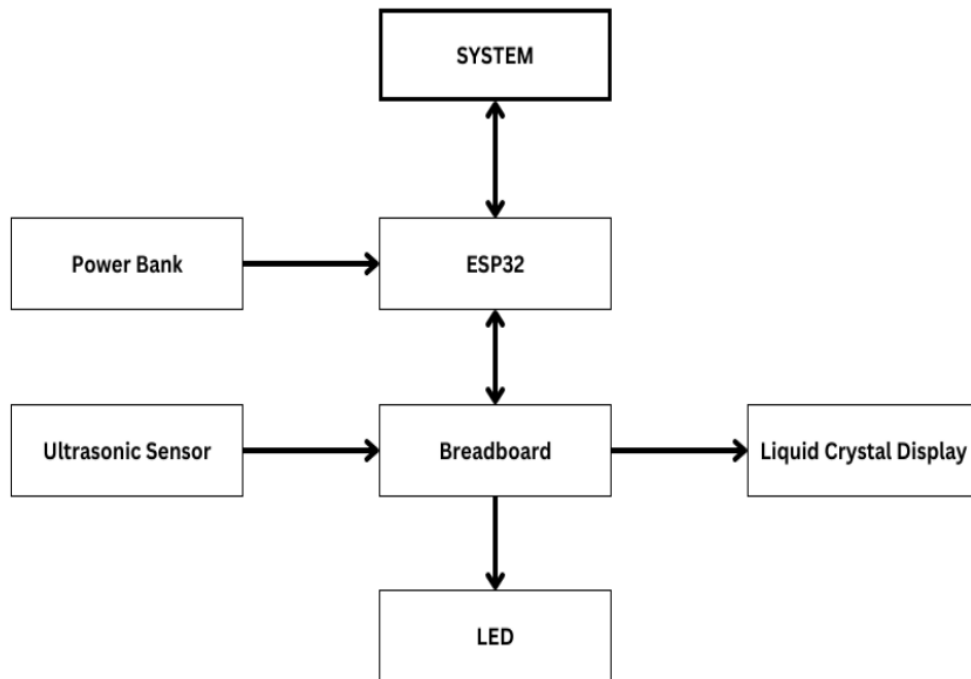


Figure 4.2.1. Block Diagram of SafeNest: Enhancing Tumaga's Emergency Response System with ESP32-Based Real-Time Flood Detection

4.2.2 Hardware Components

List of Components	Complete Name:	Specifications:
ESP32	ESP32 WROOM-32	Connectivity Technology <i>Bluetooth, Wi-Fi, USB</i> Included Components <i>3 x ESP32 CP2012 USB-C (Type-C) Development Board Module</i> Wireless Communication Standard <i>Bluetooth</i> Processor Count <i>2</i>
Breadboard	Breadboard 830 Ties	Compatible Devices <i>Personal Computer</i> Platform <i>DOS</i> Product Dimensions <i>6.5" L x 2.2" W x 0.3" H</i> System Bus Standard Supported <i>SATA 2</i> Manufacturer <i>BusBoard Prototype Systems</i> UPC <i>610074522652</i>
LED	Light Emitting Diodes (LEDs)	Light Type <i>LED</i> Special Feature <i>Long Working Time</i> Bulb Shape Size <i>Round head 3mm</i> Specific Uses for Product <i>Lamp/Arduino</i>
Power Bank	Power Bank	Connector Type <i>USB Type C, USB Type A</i> Brand <i>Mophie</i> Battery Capacity <i>10000</i> Color <i>Black</i> Special Feature <i>Fast Charging, Portable, Travel</i>
Liquid Crystal Display	I2C IIC 1602 LCD	Connectivity Technology <i>I2C</i> Included Components <i>1 x 10pin jumper wires, 3 x I2C 1602 LCD</i>

		Wireless Communication Standard <i>802.11b</i> Compatible Devices <i>Arduino, Raspberry Pi, Tinkerboard, Nano pi, Banana pi, stm32, esp32</i>
Ultrasonic Sensor	Ultrasonic Sensor HC-SR04	Included Components <i>HC-SR04</i> Wireless Communication Standard <i>Infrared</i> Compatible Devices <i>GH-311 anti-theft module</i> Processor Count <i>1</i>

4.2.3 Circuit Diagram

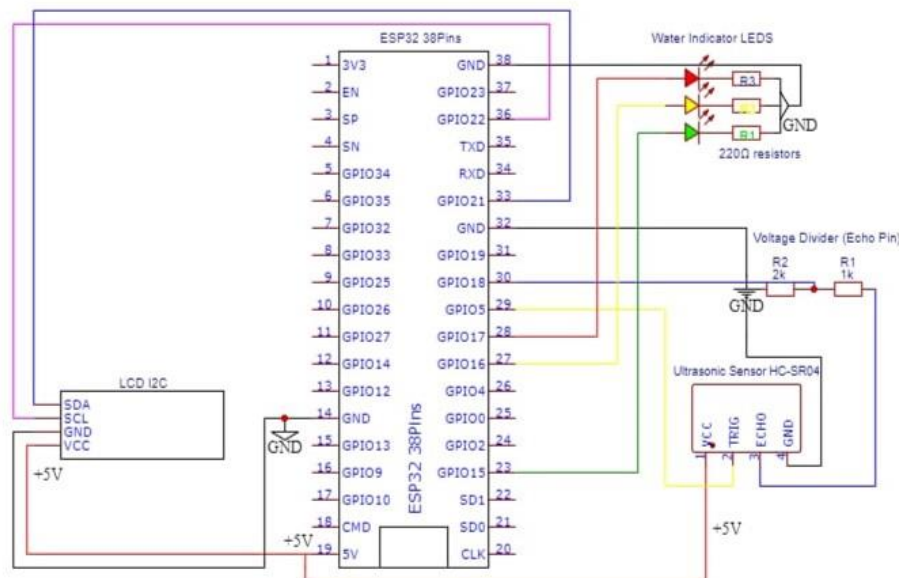


Figure 4.2.3. Circuit Diagram of Water Level using ESP32 Microcontroller

The diagram above is a circuit of the components and its connection used to develop the system. The ESP32 microcontroller with integrated Wi-Fi and Bluetooth capabilities has 38 pins with the following connections:

- Power Supply (5V)
 - Pin 19 (5V) provides 5V to the ultrasonic sensor and the LCD.
 - Ground (GND) connections are made at various points in the circuit to ensure a common reference voltage.

- GPIO Pins
 - The GPIO pins control different parts of the system, including the LEDs, ultrasonic sensor, and LCD. It acts as digital input/output (I/O) pins for communication with external devices.
- Water Indicator LEDs where each LED has a current-limiting resistor (220Ω) to protect the LEDs from drawing excessive current.
 - Red LED (High Water Level) connected to the GPIO23.
 - Yellow LED (Medium Water Level) connected to GPIO22.
 - Green LED (Low Water Level) connected to GPIO31.

Ultrasonic Sensor (HC-SR204) is responsible for measuring the distance between itself and the water surface. The working principle involves sending out an ultrasonic pulse (via the TRIG pin) and measuring the time taken for the echo to return (via the ECHO pin), which is proportional to the distance to the water.

 - VCC is connected to the 5V power supply.
 - GND is connected to ground.
 - TRIG Pin is connected to GPIO5 of the ESP32 that triggers the ultrasonic pulse.
 - ECHO Pin is connected to a voltage divider (R1 and R2) and then to GPIO18 of the ESP32. Since the ESP32 operates at 3.3V, a voltage divider ($2k\Omega$ and $1k\Omega$ resistors) is used to step down the 5V signal from the sensor to 3.3V, protecting the ESP32 input pin.
- LCD (I2C Interface) for displaying information such as the water level. The LCD is connected via an I2C interface and also has power connections such: VCC (5V) and GND, which requires only two wires for communication:
 - SDA (Data Line) is connected to GPIO21.
 - SCL (Clock Line) is connected to GPIO22.
- Power Supply using a 5V supply, which provides power to the ESP32 (via Pin 19), the ultrasonic sensor, and the LCD display.

The ESP32 acts as the central controller, reading sensor data from the Ultrasonic Sensor HC-SR04 and using that data to determine the water level. Based on the water level, the ESP32 activates different LEDs to visually indicate whether the water is low, medium, or high. Additionally, the LCD can be used to display the exact water level in real-time. The voltage divider is an important safety measure, ensuring that the ESP32 does not receive higher voltage than it can handle from the ultrasonic sensor's echo pin.

4.2.4 Testing the Ultrasonic Module

The objective of this test is to verify the accuracy and performance of the ultrasonic sensor in measuring water levels in real-time and displaying the data on a dashboard.

Components:

- Ultrasonic Sensor (HC-SR04)
- ESP32 Microcontroller
- LCD Display (1602)
- 5V Power Supply
- Water Container (e.g., aquarium or bucket)
- Firebase for real-time data logging
- Wi-Fi connection for data transfer

4.2.4.1 Test Procedure

The components and hardware devices used to measure the data of the ultrasonic sensor and test its accuracy include the ultrasonic sensor (HC-SR04), ESP32 microcontroller, LCD display (1602), 5V power supply, and a small water container to simulate changing water levels. The sensor was calibrated by positioning it 32 cm from the bottom of the water container. The test was conducted in an indoor environment, where the water container simulated rising and falling water levels, allowing for accurate and controlled measurements.

4.2.4.2 Hardware Setup

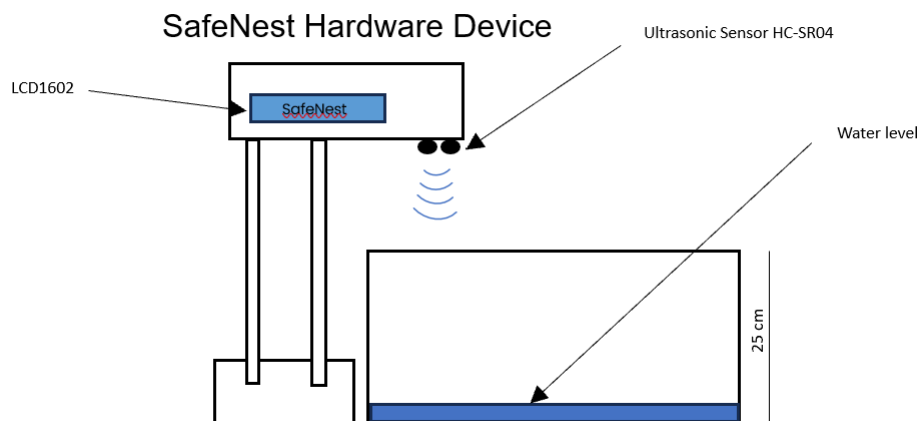


Figure 4.2.4.2 Hardware Setup

Mount the ultrasonic sensor above the water container, positioning it 32 cm from the bottom, and connect it to the ESP32 and LCD to display live water level data. The ultrasonic

sensor emits ultrasonic waves or pulses. Specifically, it generates high-frequency sound waves (typically above 20 kHz, which is beyond human hearing range) and then listens for the echo that bounces back after the waves hit an object or in this particular scenario it waits if the water level is rising or decreasing. The sensor measures the time it takes for the echo to return, which is used to calculate the distance to the water level.

The ESP32 was set up to send real-time data to Firebase where dynamic data is going store the water level data is the most important among other data simply because it sends the raw data then the system will filter the only important data.

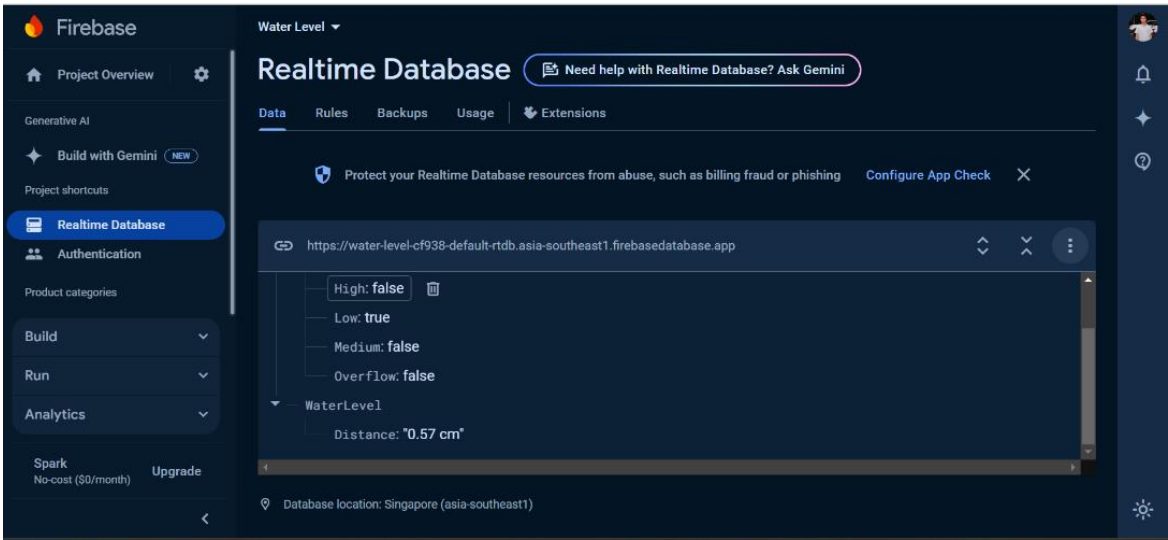


Figure 4.2.4.2 Firebase Database

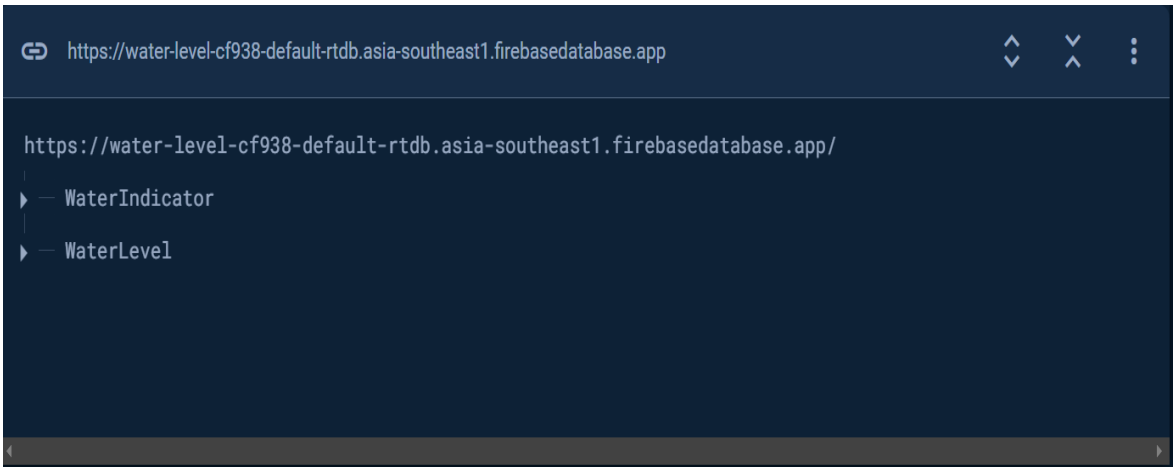


Figure 4.2.4.2 Firebase Database Data (e.g Water Indicator and Water

4.2.4.3 Software Setup

The software development for the *SafeNest: Contactless Based Real-time Flood Detection and Monitoring System* utilizes a combination of tools and technologies to configure, collect, and display real-time water level data. The primary tools used are the Arduino IDE for configuring the ESP32 microcontroller and sensor data, and Visual Studio Code (VS Code) for developing the dashboard and login page.

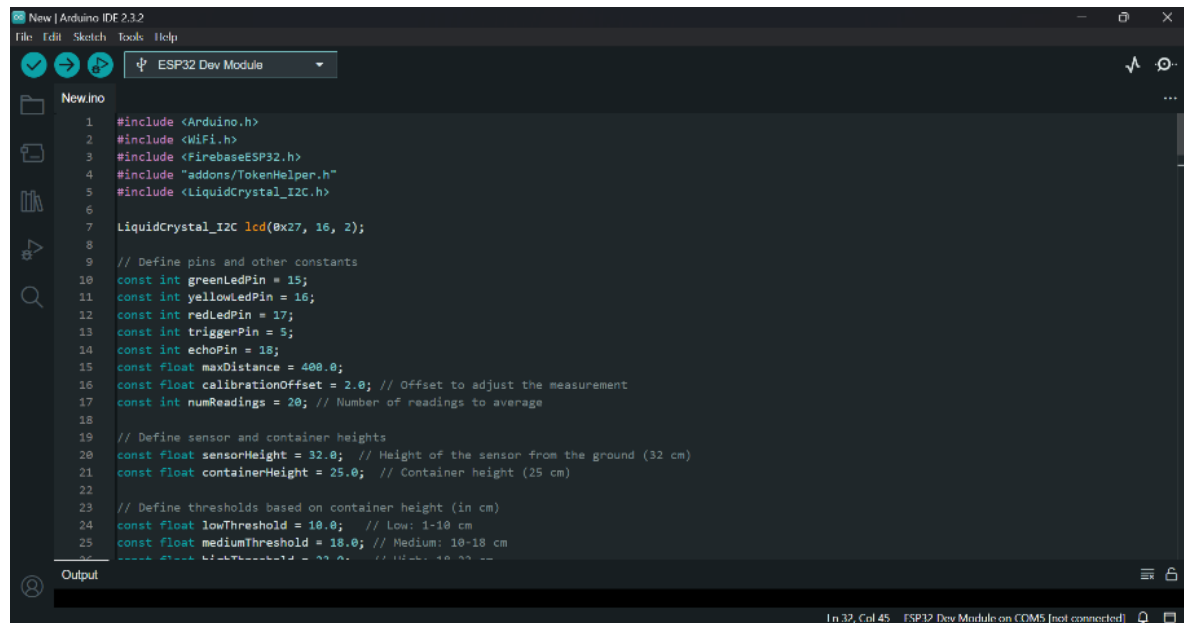


Figure 4.2.4.3 Arduino IDE

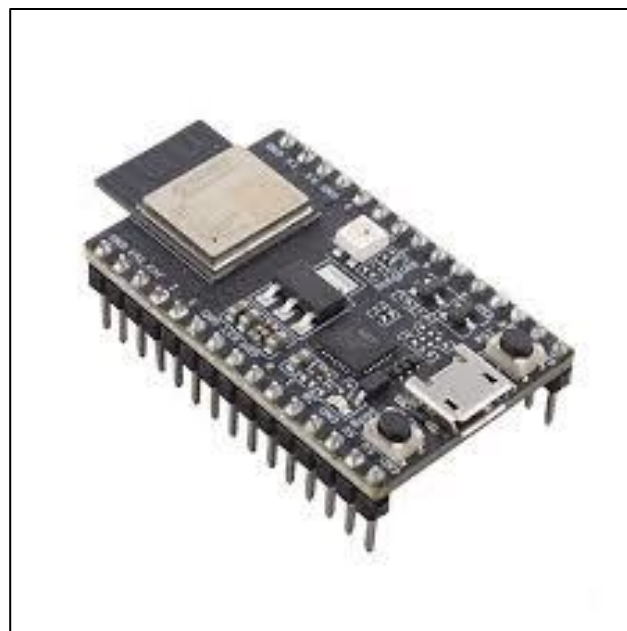


Figure 4.2.4.3 ESP32 - WROOM

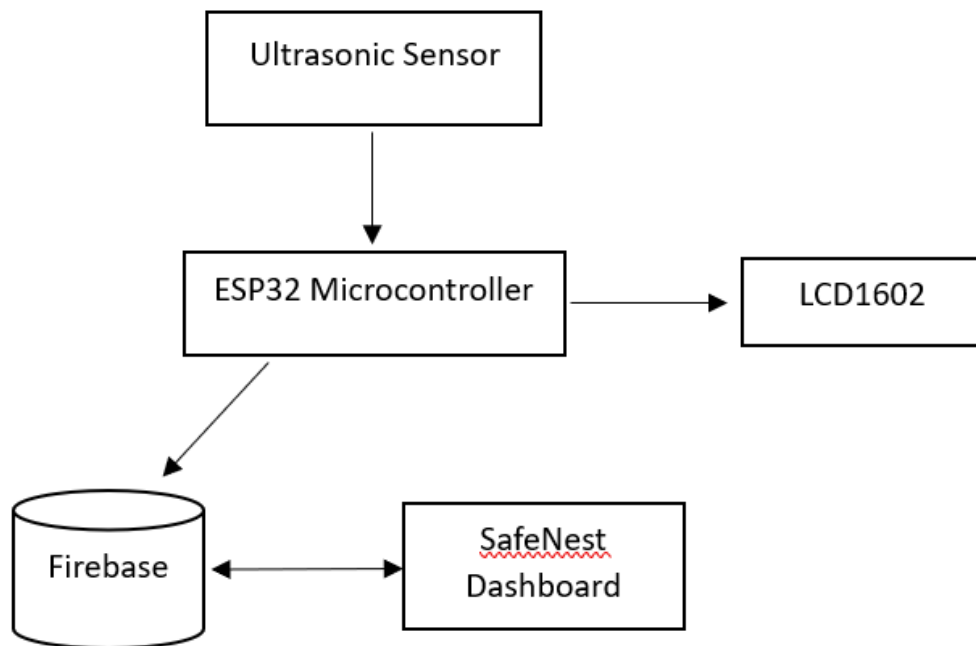


Figure 4.2.4.3 Block Diagram – Data Transferred

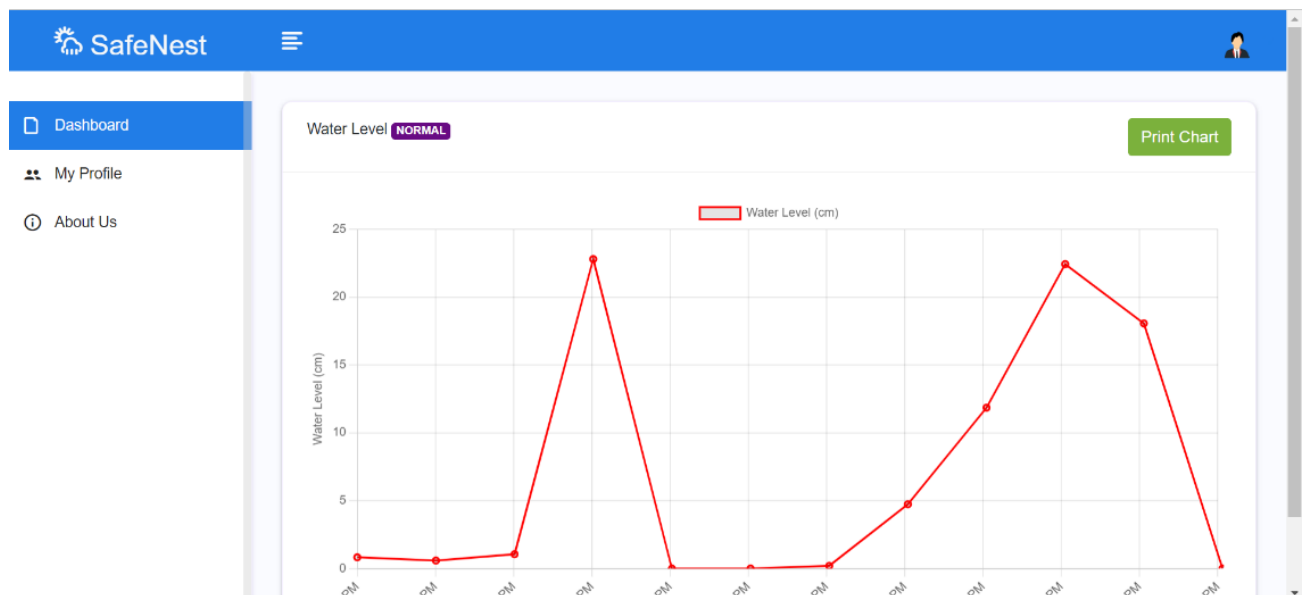


Figure 4.2.4.3 SafeNest Dashboard

The ESP32 microcontroller collects distance data from the ultrasonic sensor (HC-SR04) to calculate the water level. This calculated water level data is then sent to a Firebase database in real-time. The Firebase database acts as a cloud storage solution, allowing the data

to be retrieved easily. From Firebase, the real-time water level data is pulled into a dashboard, where it is visually displayed for monitoring purposes. The dashboard categorizes the water levels into predefined thresholds—Low, Medium, High, and Overflow—providing users with clear and actionable insights into the water level status.

4.3 Software Development Life Cycle

The utilization of Agile methodology for monitoring water levels at the riverbank of Barangay Tumaga provides notable advantages due to its flexibility, iterative development approach, and emphasis on collaboration. Agile allows for continuous feedback from stakeholders, ensuring the project can evolve based on actual needs and challenges identified throughout its development. The adaptability promotes faster problem-solving and the integration of new technologies or techniques as it emerges. Key benefits include enhanced engagement with stakeholders, the capacity to pivot quickly in response to data and insights, incremental delivery of working components, and improved team collaboration, all of which contribute to a more effective and responsive project outcome in an academic environment.

4.3.1 System Requirements

The system requires IoT sensors for water level measurement, a reliable data transmission method, a cloud or local database for storage, and a user interface for monitoring and alerts.

4.3.2 Requirements Analysis

4.3.2.1 Use Case Diagram

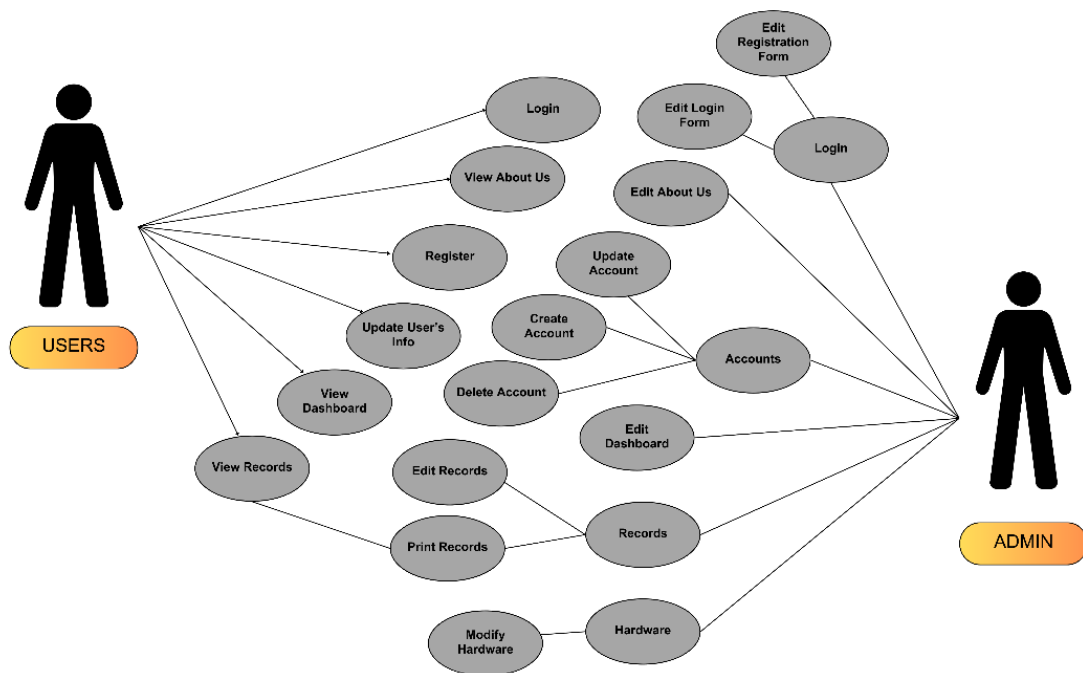


Figure 4.3.2.1 Use Case Diagram of the Software System

4.3.2.2 Use Case Specification

- User Case 1: Register
 - User: Administrator
 - Description: The system must allow the user to create an account on the system.
 - Flow: The user inputs all the necessary information to create the account.
 1. Create Account form is loaded.
 2. Fill in all the fields: the first name, last name, email, number, password, and confirm password. When done, click "Sign up" button.
 3. A message box will prompt "Admin Created Account," click "Ok"
- Use Case 2: Login
 - User: Administrator, Captain, Kagawad, SK Kagawad

- Description: The system shall allow the user to have access on the system.
- Flow: The user inputs a validated username and password, made by the admin itself; it should match to access the assigned module after login.
 1. Login form is loaded
 2. Enter username and password.
 3. If the login button is clicked, the system will verify the inputted values. If one of the values are invalid, a message will prompt “Incorrect email or password!” If the entered values are correct, a message box will prompt “Login Successfully.”
- Use Case 3: View Dashboard
 - User: Administrator
 - Description: The system allows the user to view the dashboard module.
 - Flow: The user clicks the “Dashboard Navigation Menu” which allows to view the dashboard interface.
 1. Water Status displays the water status that includes: low, normal and high, indicating the current water level.
 2. When clicked, the Water Status button send an email that triggers when water level reaches a certain level that allows to manually send an email notification.
 3. Graphical Status button indicates the graphical update for timely, monthly, and yearly logs; implying that the system is reporting data promptly. When clicked, it directs to the Records for the system’s logs.
 4. Users Count displays the number of users in the system.
 5. Main graph is visible that tracks water level; with the Y-axis labeled “Water Level (cm)” and the X-axis labeled the “Time”. A legend above the chart

for color reference that corresponds to the red dot on the graph.

6. Print Chart button that allows the user to print or save the graphical data presented.

- Use Case 4: View Records

- User: Administrator, Role Members
- Description: The system allows to tracks user activity and events on the system.
- Flow: The user clicks the “Records Navigation Menu” which allows to view the records interface.
 1. System’ Log is shown where it contains a table of user’s activity such as: logins, logouts, and account creation, email notification, and its timestamp; pagination with navigation buttons to scroll through the log entries and a button to export the log data in various formats that includes: Copy, CSV, and PDF file for record-keeping
 2. Search input field that allows the user to filter log entries, and search specific activities making it easier to find specific activities based on keywords.

- Use Case 5: View User Information

- User: Administrator, Role Members
- Description: The system allows the user to view detailed information about personnel in the system.
- Flow: The user clicks the “User Info Navigation Menu” which allows to view the User Info interface.
 1. Personnel Overview accesses by the user the Personnel section which is visible represented by a card with the name, email address, an option for view more detailed information.
 2. Search and Add Functionality that has the (3) three action buttons: Add personnel, Roles, and Legend.

3. Personnel Information Display contains a graphic for easy identification, email address and name.
4. View Profile when clicked, is taken to a more detailed page with additional information about the selected person.

- Use Case 6: View About Us
 - User: Administrator, Role Members
 - Description: The system allows the user to view detailed information about developers of the system.
 - Flow: The user clicks the “About Us Navigation Menu” which allows to view the About Us interface for the displayed information of the developers.

4.3.3 Functional Requirements

The system allows user authentication, shall provide real-time data updates from sensors, it should send alerts for notifying a certain threshold reached, and to have access for environmental data.

4.3.3.1 User Registration

- The system provides a registration form for administrators to create user accounts.
- It must validate input fields such as first name, last name, email, phone number, and password before allowing account creation.
- Upon successful registration, the system must display a confirmation message.

4.3.3.2 User Authentication

- The system allows users (Administrator, Captain, Kagawad, SK Kagawad) to log in with a validated username and password.
- It must verify the credentials against stored values and display error messages for invalid attempts.
- On successful login, a success message must be displayed, granting access to the user’s assigned module.

4.3.3.3 Dashboard Access

- The system must provide a dashboard that displays real-time water level status, indicating low, normal, and high levels.
- It must allow users to send email notifications manually when water levels reach critical levels.
- A graphical representation of water levels over time (timely, monthly, yearly) must be available on the dashboard.
- The total number of system users must be displayed.
- The dashboard must include an option to print or save graphical data for record-keeping.

4.3.3.4 Record Management

- The system tracks user activities such as logins, logouts, and account creation, and maintain a system log.
- Pagination must be supported to navigate log entries, and the system must allow exporting of logs in formats like CSV, PDF, or copy to clipboard.
- A search functionality must be available to filter log entries and retrieve specific activities.

4.3.3.5 User Information Access

- The system provides a detailed view of personnel information, including name, email, and other relevant details.
- It allows users to search for personnel, add new personnel, assign roles, and access other administrative options.
- A detailed profile view for each user must be accessible through a “View Profile” option.

4.3.3.6 About Us Section

- The system must include an “About Us” page that displays detailed information about the developers of the system and its purpose.

4.3.4 Non-Functional Requirements

- **Scalability.** The system is designed to allow future integration of additional sensors or enhancements to the communication network.
- **Performance.** The system provides accurate real-time data and ensure timely alerts without significant delay.
- **Security.** The sensitive data, such as user credentials and flood data are securely stored and transmitted to prevent unauthorized access.
- **Usability.** The system has an intuitive and user-friendly interface making it easy for users with varying technical expertise to navigate and perform tasks efficiently.
- **Reliability.** The system ensures high availability and minimal downtime to maintain continuous monitoring and timely alerts.
- **Data Integrity.** The system guarantees the accuracy and consistency of the data being processed and stored, preventing loss or corruption of data during operations.

4.3.5 Context Diagram

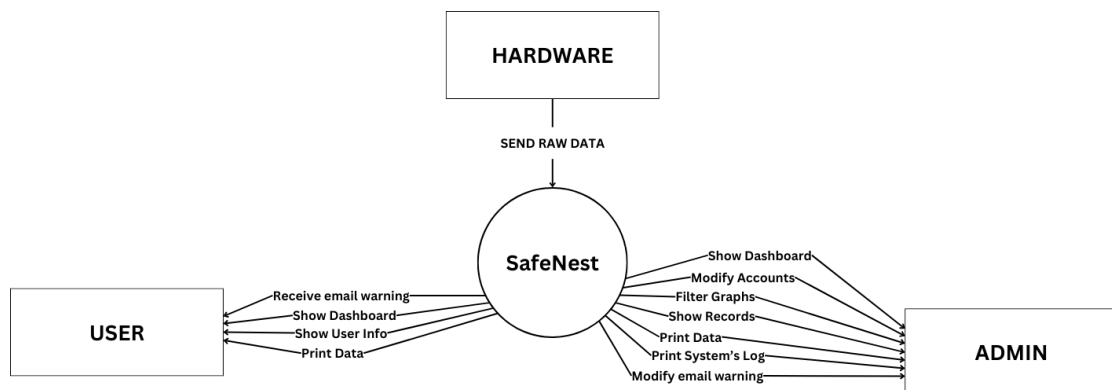


Figure 4.3.5 Context Diagram

The diagram above shows the overview of the SafeNest system and its interactions with external entities, specifically hardware, users, and administrators. The SafeNest as the central component, facilitates the flow and processing of raw data, ensuring seamless communication between the different entities. The *hardware* entity refers to physical devices or sensors that generate raw data; continuously monitor and collect information, such as environmental metrics or system status data. It is designed to send this raw data directly to SafeNest for processing. At the center, the SafeNest is responsible for receiving, processing, and managing the data. The system takes raw input from the hardware and uses it to perform specific actions that are useful for both users and administrators. SafeNest likely incorporates algorithms, data

storage, and other backend services to ensure proper data handling. The *user* who represents as the end-users who interacts the system. It typically accesses processed data and notifications generated by SafeNest. The *admin* is responsible for overseeing the system’s operations. It manages configurations, handle system updates, and ensure that the hardware, SafeNest, and user interactions are functioning properly. Administrators have privileges to modify, troubleshoot, or access detailed logs and reports from SafeNest.

4.3.6 Data Flow Diagram

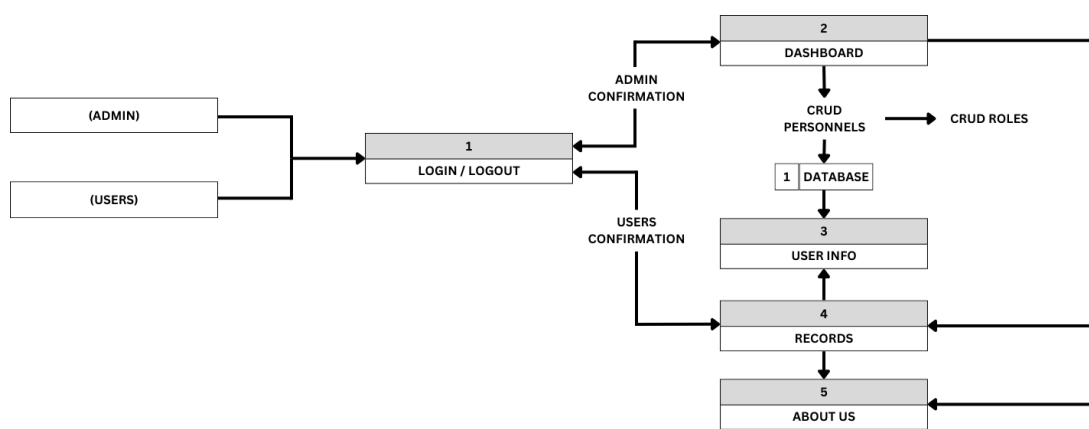


Figure 4.3.6.1 Admin Login Data Flow Diagram

The provided diagram above represents a system where both administrators and users can interact with a centralized platform, the water level monitoring system. The system enables two types of users, Admin and Users, to access the platform through a shared Login/Logout feature. Once authenticated, administrators are directed to an Admin Dashboard, which provides functionalities for managing system users, viewing user information, and maintaining records. Users requires to login but the interaction with the system is limited as users can only view the dashboard and other non-administrative data. The platform is database-driven with both user and admin operations relying on CRUD operations.

4.3.6.1 Admin Login Data Flow Diagram

The admin can login with a predetermined account exclusively for admin only.

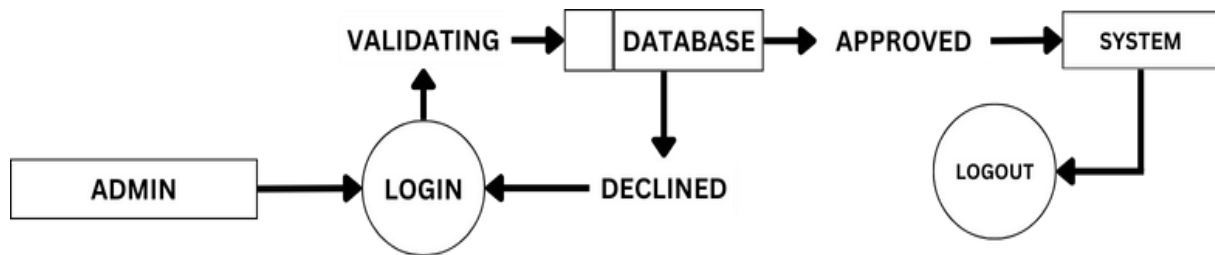


Figure 4.3.6.1 Admin Login Data Flow Diagram

4.3.6.2 User Interface Structure Data Flow Diagram

This User Interface Structure diagram provides a more detailed view of the user authentication flow for a monitoring system involving hardware components (likely for water or flood monitoring). The diagram outlines the processes that occur when a user logs in, the validation of their credentials, and the post-login processes, such as accessing dashboards, user information, raw data, and profile management.

The system integrates a database, which stores user credentials, roles, and hardware data. There is also a validation process that determines whether a user is granted or denied access based on the login attempt.

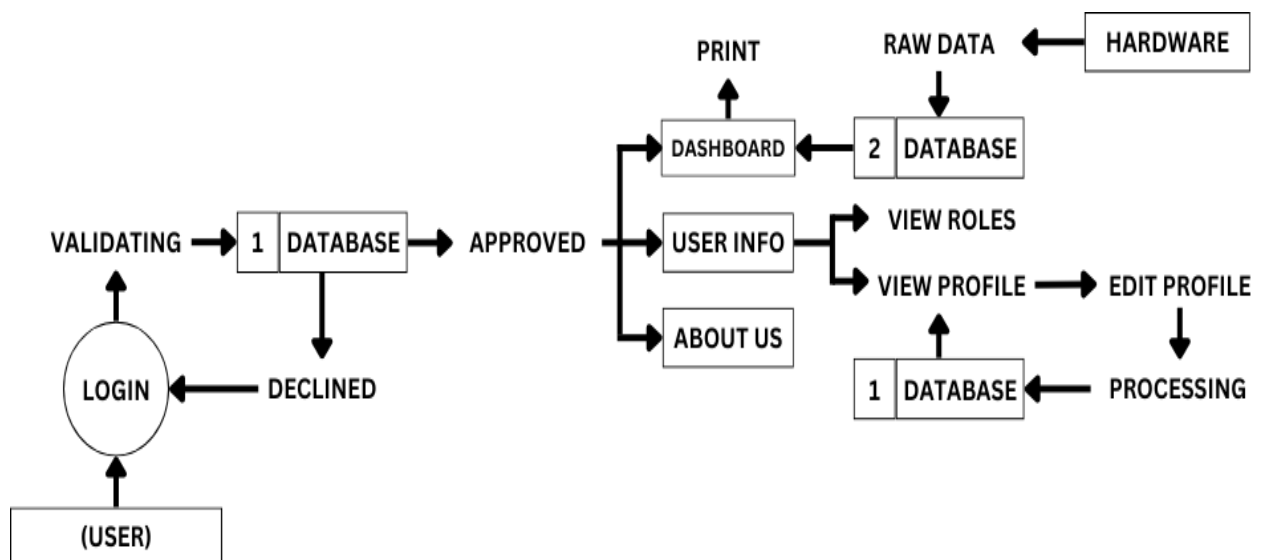
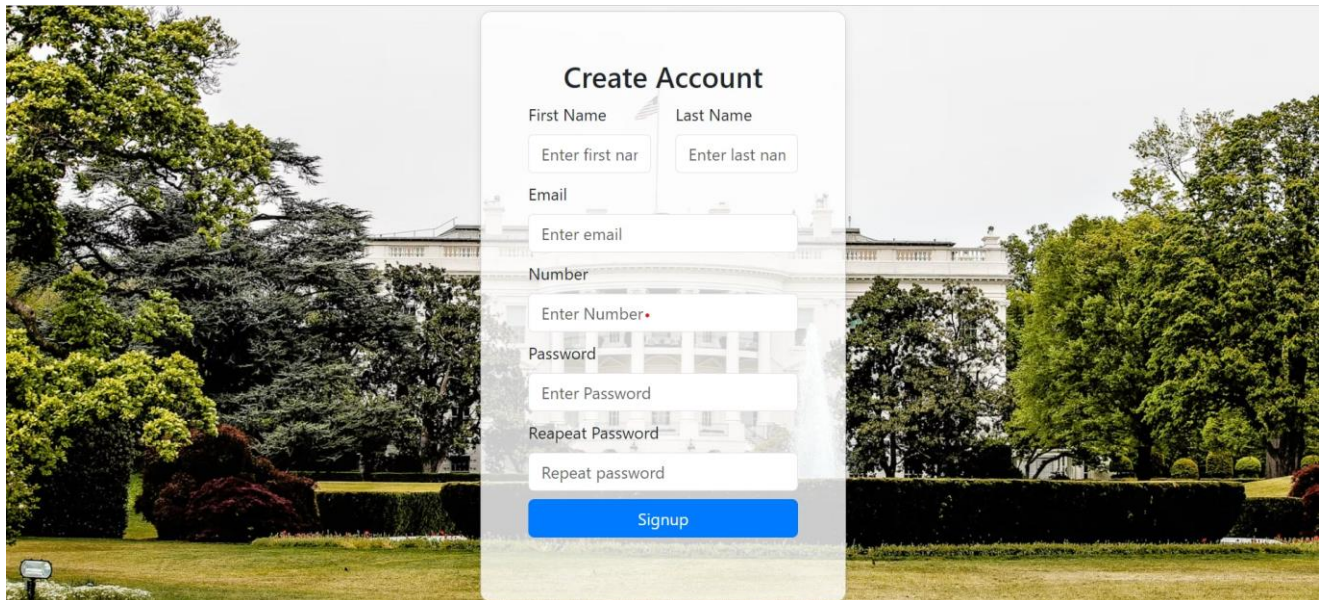


Figure 4.3.6.2 User Interface Structure Data Flow Diagram

4.3.7 User Interface Design

4.3.7.1 Web Application View – Admin

This feature allows the admin to create an account to gain access to the application.



The image shows a 'Create Account' form overlay on a background image of the White House. The form includes fields for First Name, Last Name, Email, Number, Password, and Repeat Password, along with a Signup button.

Create Account

First Name

Last Name

Email

Number

Password

Repeat Password

Figure 4.3.7.1.1 Web View Login

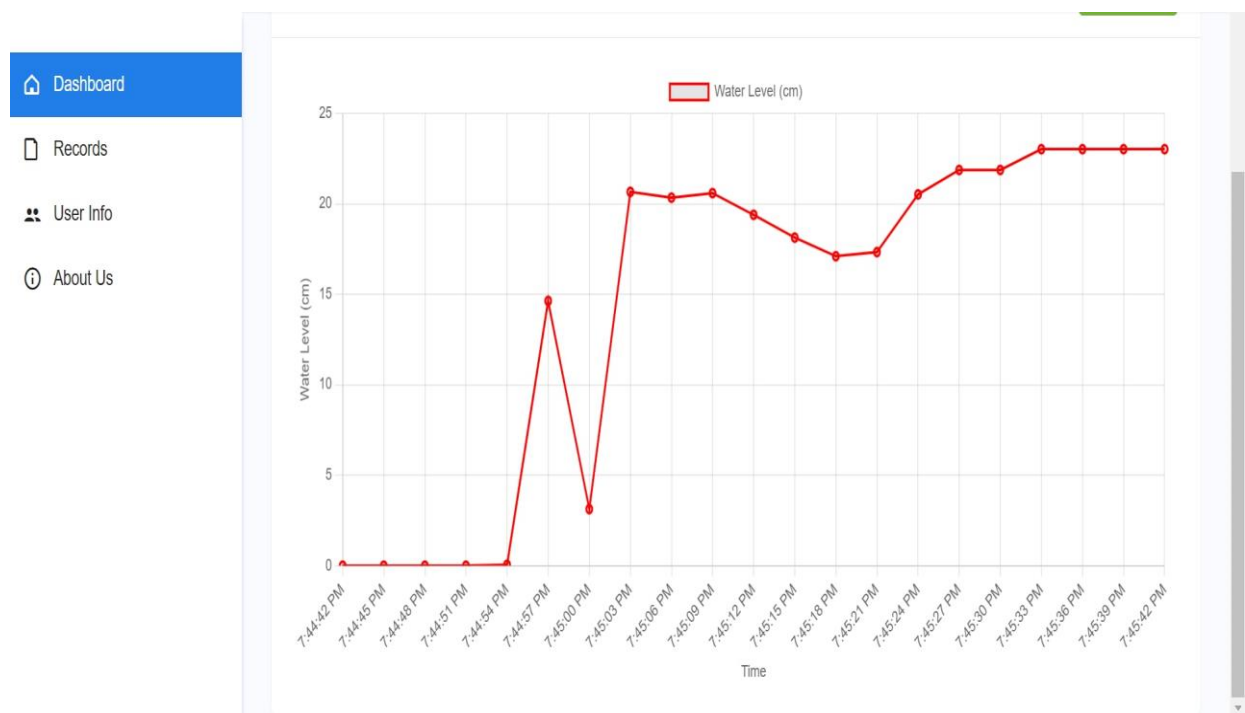


Figure 4.3.7.1.2 Web View – Dashboard Admin

This feature gives the admin an overview of the dashboard.



Figure 4.3.7.1.3 Web View – Dashboard Admin

This feature gives the admin an overview of the dashboard to save and print the chart and the filtering option for the time interval displaying of the water level.

#	Time	Activity
#	2024-09-29 19:07:59	Brandon Kyle Rojas is logged in
#	2024-09-29 19:07:48	Brandon Kyle Rojas is Created an Account and set roles as Admin

Showing 1 to 2 of 2 entries

Figure 4.3.7.1.4 Web View – Record Admin

This feature gives the admin the access to view the system's log in records navigation menu.

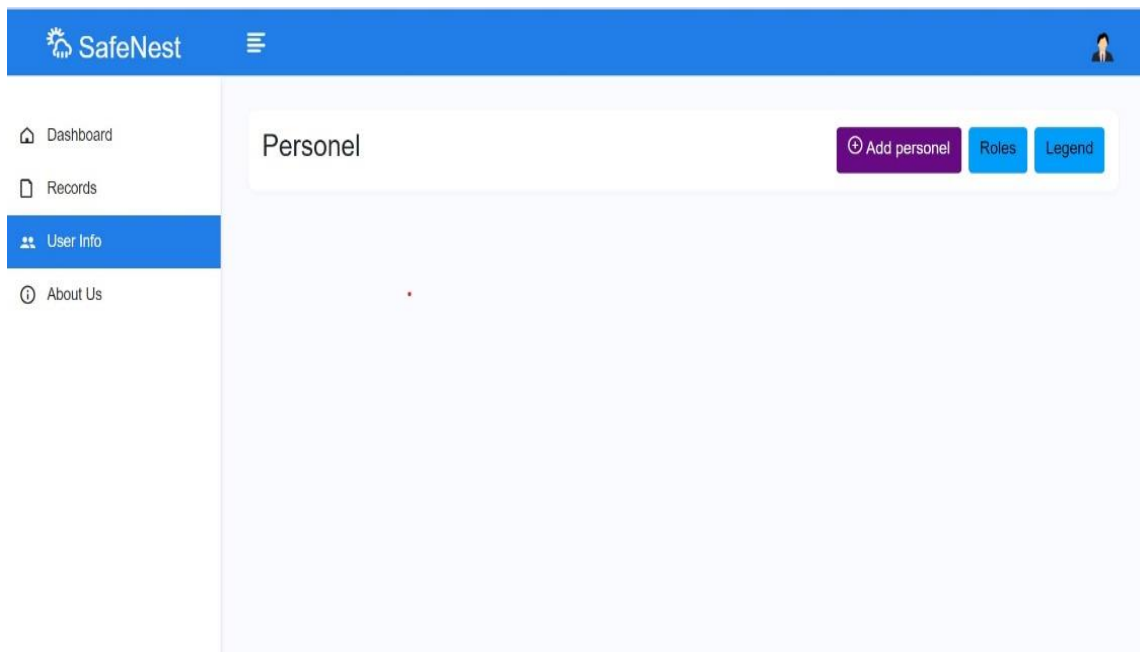


Figure 4.3.7.1.5 Web View – User Info Admin

This feature gives the admin to add personnel, add roles, and to view the legend of the users created.

Figure 4.3.7.1.6 Web View – User Info Add Personnel Admin

The admin can add the credentials needed to add a user in the system.

Roles

Add personnel
Roles

Add Roles

Roles name

Color designation

Submit

#	Roles	Actions
1	Barangay Captain	Edit Delete
2	Kagawad	Edit Delete
3	SK Kagawad	Edit Delete

Figure 4.3.7.1.7 Web View – User Info Roles Admin

The admin can edit the user’s roles or to delete the user.

Roles Color Coding

Color

Barangay Captain

Color

Kagawad

Color

SK Kagawad

Close

Figure 4.3.7.1.8 Web View – User Info Legend Admin

The admin views the specific legends created for the users in the system.

4.3.7.2 Web Application View – User

This feature allows the user to gain access to the system. All user-level can access the application provided they have their username and password created.

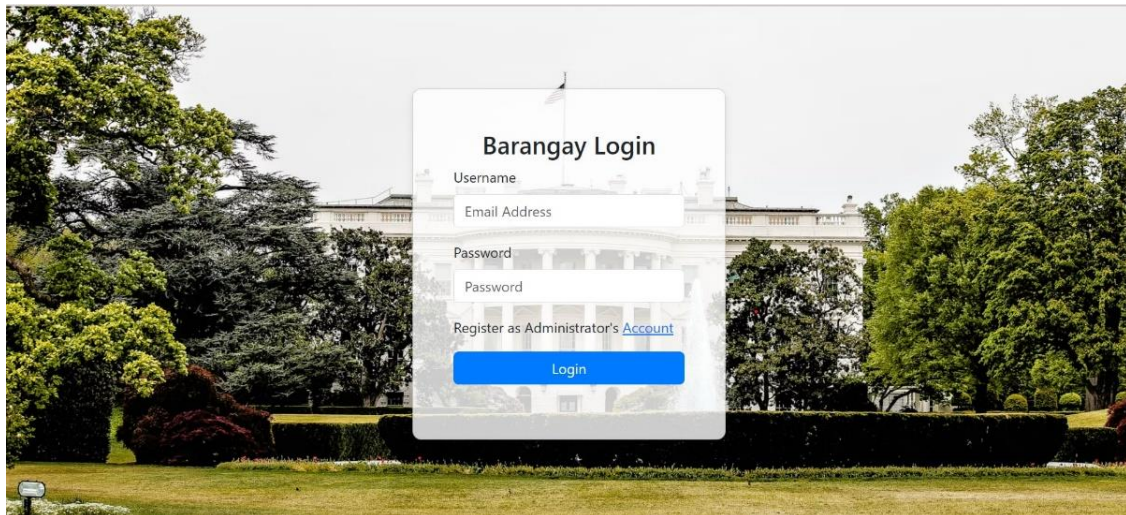


Figure 4.3.7.2.1 Web View Login - User

Figure 4.3.7.2.2 Web View My Profile – User

The user can edit the profile created by the admin.

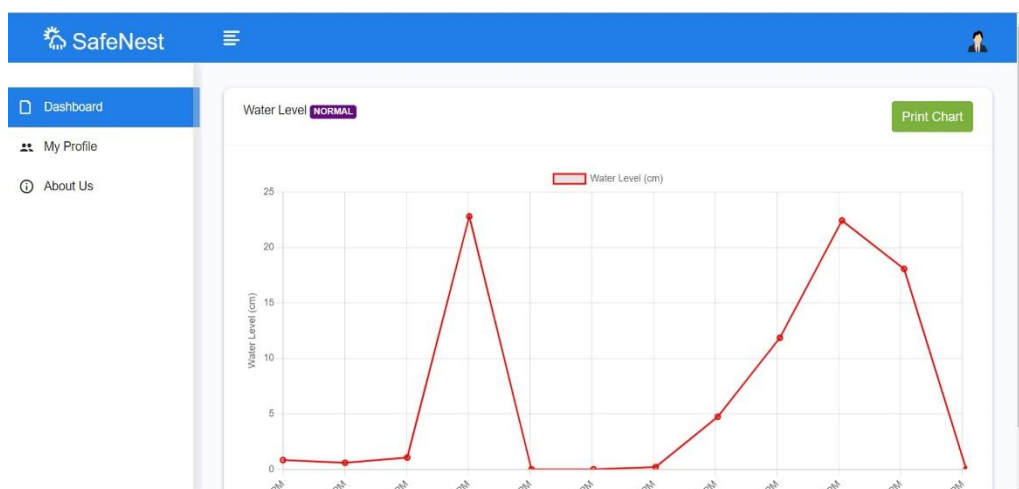


Figure 4.3.7.2.3 Web View Dashboard – User

The user views the dashboard and can save as pdf or print the graphic chart.

4.3.8 State Chart Diagram

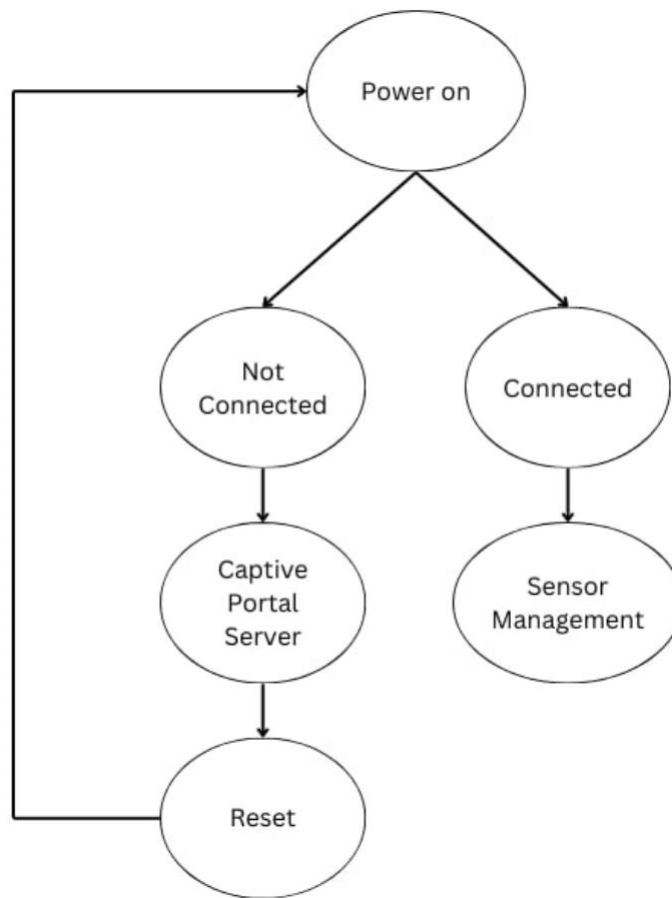


Figure 4.3.8 State Chart Diagram

The state chart diagram for the captive portal system begins with the "Power On" state, transitioning into two possible paths: "Not Connected" or "Connected to Wi-Fi." If the system is not connected, it moves to the "Captive Portal Server" state, where users can configure the network settings, with an option to reset and restart the process. Once connected to Wi-Fi, the system advances to the "Sensor Management" state to perform its primary operations, which run in a continuous loop. It initializes a cycle counter and processes server handling, Wi-Fi monitoring, and periodic delays of 200 milliseconds. If the cycle counter reaches 3, representing one minute, the system verifies Wi-Fi connectivity and collects sensor data before resetting the cycle, ensuring consistent and efficient operation.

4.3.9 Flowchart

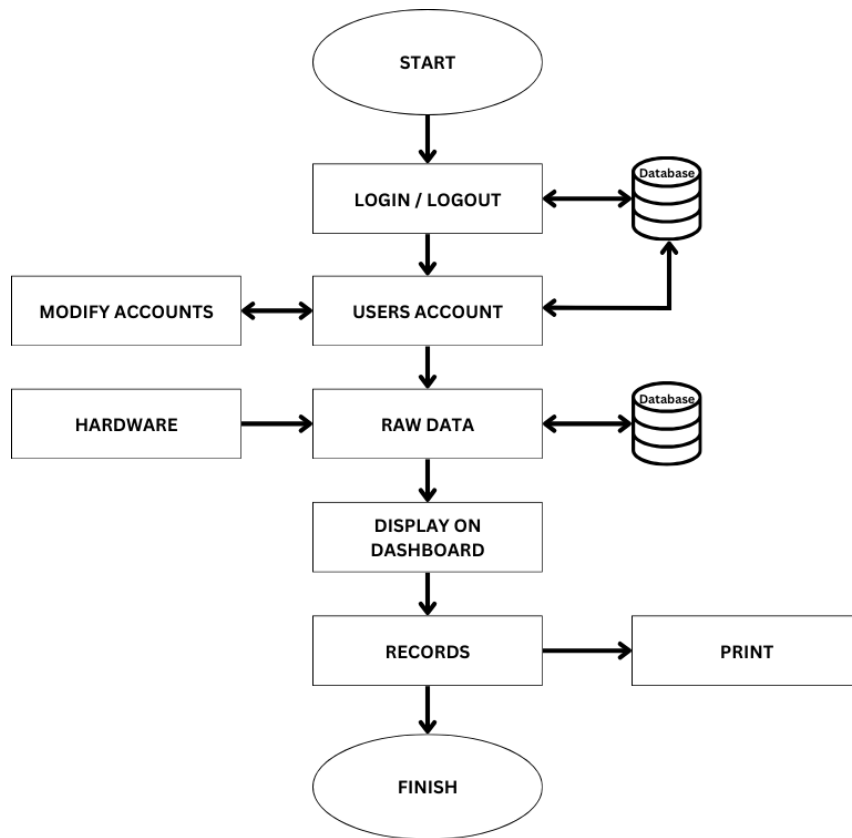


Figure 4.3.9 Flowchart

The flowchart above outlines the operations of a system such as one used for water level or flood monitoring. The process starts with user authentication, followed by interaction with user accounts and hardware, then proceeds to data display and record management, before finishing with an optional print functionality.

4.3.10 Agile Development

The development process of the flood monitoring system followed an Agile methodology, which allowed for flexibility, iterative progress, and continual stakeholder feedback. This methodology was particularly well-suited to the dynamic nature of the project, as it provided the capability to adapt to changes, address real-time feedback, and refine system components through multiple development cycles, known as sprints. The total project timeline spans approximately 25 weeks (6 months) that follows a 2-week sprint cycle with time allocations for testing, feedback, and refinements

4.3.10.1 Sprint 0: Concept and Vision (Week 1)

Define the project's objectives, stakeholders, and main functionalities. To develop the initial roadmap.

4.3.10.2 Sprint Iterations 1-12: Hardware and Software Development

4.3.10.2.1 Sprint 1: Hardware Setup and Sensor Integration (Week 2-3)

In the first sprint, the focus was on integrating a functional prototype for water level measurement. Setting up the ESP32 microcontroller and connect the HC-204 ultrasonic sensor for accurate water level detection. Develop a basic code for water level measurement and calibration; to test the sensor readings in various conditions.

4.3.10.2.2 Sprint 2: Software – Firebase Setup (Week 4-5)

The second sprint is to configure firebase for data storage of water level data. Setting up the Firebase for real-time database and testing the basic data transmission from ESP32 to Firebase. Creating initial database structure for water level readings.

4.3.10.2.3 Sprint 3: Hardware - LED and LCD Display Functionality (Week 6-7)

The third sprint, the goal is for visual representation of water levels. Implementing LED indicators (low, medium, high levels). To configure the LCD1602 (I2C) for displaying water level percentages. Testing the hardware synchronization (sensor, LEDs, and LCD).

4.3.10.2.4 Sprint 4: Software – Backend Development (Week 8-9)

The fourth sprint, to create a backend structure for data handling. Develop Firebase functions to fetch data and ensures security and scalability for data storage. Integrating backend with Firebase for real-time updates. The backend system ready to serve data to the dashboard.

4.3.10.2.5 Sprint 5: Hardware Optimization (Week 10-11)

The fifth sprint is to test and optimize hardware reliability. Testing the sensor accuracy under various conditions. Optimizing ESP32 code for stable operation and addressing hardware issues, the power supply and internet connection)

4.3.10.2.6 Sprint 6: Frontend Dashboard Development (Week 12-13)

The sixth sprint is to build the web dashboard for data visualization, to design the user interface using HTML, CSS, and JavaScript. Integrating real-time data from Firebase and implement features such as historical data and graphs.

4.3.10.2.7 Sprint 7: Mock Defense Deployment Preparation (Week 14-15)

The seventh sprint is preparing for hardware and software functionalities for deployment. Building durable enclosures for hardware, testing power efficiency and document deployment setup.

4.3.10.2.8 Sprint 8: System Integration Testing (Week 16-17)

The eighth sprint is to test the entire system for end-to-end functionality. Integrating hardware with the Firebase backend and dashboard; testing the flow of data from sensor to the dashboard and fixing bugs, inconsistencies, and feedback coming from the result of the panelist.

4.3.10.2.9 Sprint 9: Role-Based Dashboard Features (Week 18-19)

The ninth sprint is adding user roles and permissions to the dashboard; implementing authentication for admin and end-user roles. Add features like alert as sending SMS notification and ensuring end-users only relevant data.

4.3.10.2.10 Sprint 10: Captive Portal and Twilio SMS (Week 20-21)

The tenth sprint is creating a Wi-Fi login page for local access were configuring ESP32 to host a captive portal; allowing user to access system water level data via Wi-Fi and testing portal usability and functionality.

4.3.10.2.11 Sprint 11: SMS Notification using Twilio (Week 22-23)

The eleventh sprint is implementing SMS alerts for critical water levels by integrating Twilio API for sending SMS notifications. Configure triggers for sending alerts when water level reaches at high and testing SMS delivery and system reliability.

4.3.10.2.12 Deployment and Final Adjustments (Week 24-25)

The twelve sprint is to deploy the system and gather user feedback; presenting the system to the panelist and collecting feedback and plan improvements.

4.3.11 IoT Device Integration

The integration of IoT devices was critical to achieving the real-time monitoring and flood detection objectives of the project. The ESP32 microcontroller was chosen as the central hub for data processing and communication due to its low power consumption, integrated Wi-Fi capability, and versatility in handling multiple sensors.

4.3.11.1 Ultrasonic Sensors

The system utilized ultrasonic sensors for measuring water levels. The sensors were placed at strategic points along the riverbank in Barangay Tumaga to continuously monitor water depth. The ultrasonic sensors measured the distance between the water surface and the sensor, and this data was sent to the ESP32 for real-time analysis. The sensors were programmed to trigger alerts at three threshold levels: low, medium, and high. These thresholds were calibrated through multiple tests runs to ensure accurate flood risk assessments.

4.3.11.2 ESP32 Micro-controller

The ESP32 acted as the core processing unit of the system. It collects data from the ultrasonic sensors, processed it, and communicates the information to the central server. When water levels reached the critical threshold, the ESP32 initiated a sequence of events, including activating LED indicators, displaying information on an LCD, and sending out SMS alerts via the Twilio API. The ESP32 was connected to the local Wi-Fi network, allowing for seamless data transmission to the cloud-based Firebase system for real-time monitoring.

4.3.11.3 Communication and Alerts

The system was designed to alert local barangay officials of impending floods. When the ultrasonic sensors detected high water levels, the ESP32 triggered an alert mechanism that sent SMS notifications to registered users, including barangay officials and response teams. It achieves through an integration with the Twilio API, which enabled automated text message alerts whenever water levels surpassed critical limits.

4.3.11.4 Dashboard and Data Storage

A web-based dashboard was developed to display real-time water level data, historical records, and generate flood risk analysis. The dashboard was built using Firebase for storing and retrieving real-time sensor data, while MySQL was used for managing user accounts and historical records. The dashboard allowed users to visualize water levels, review historical trends, and export reports for disaster planning.

4.3.12 Deployment

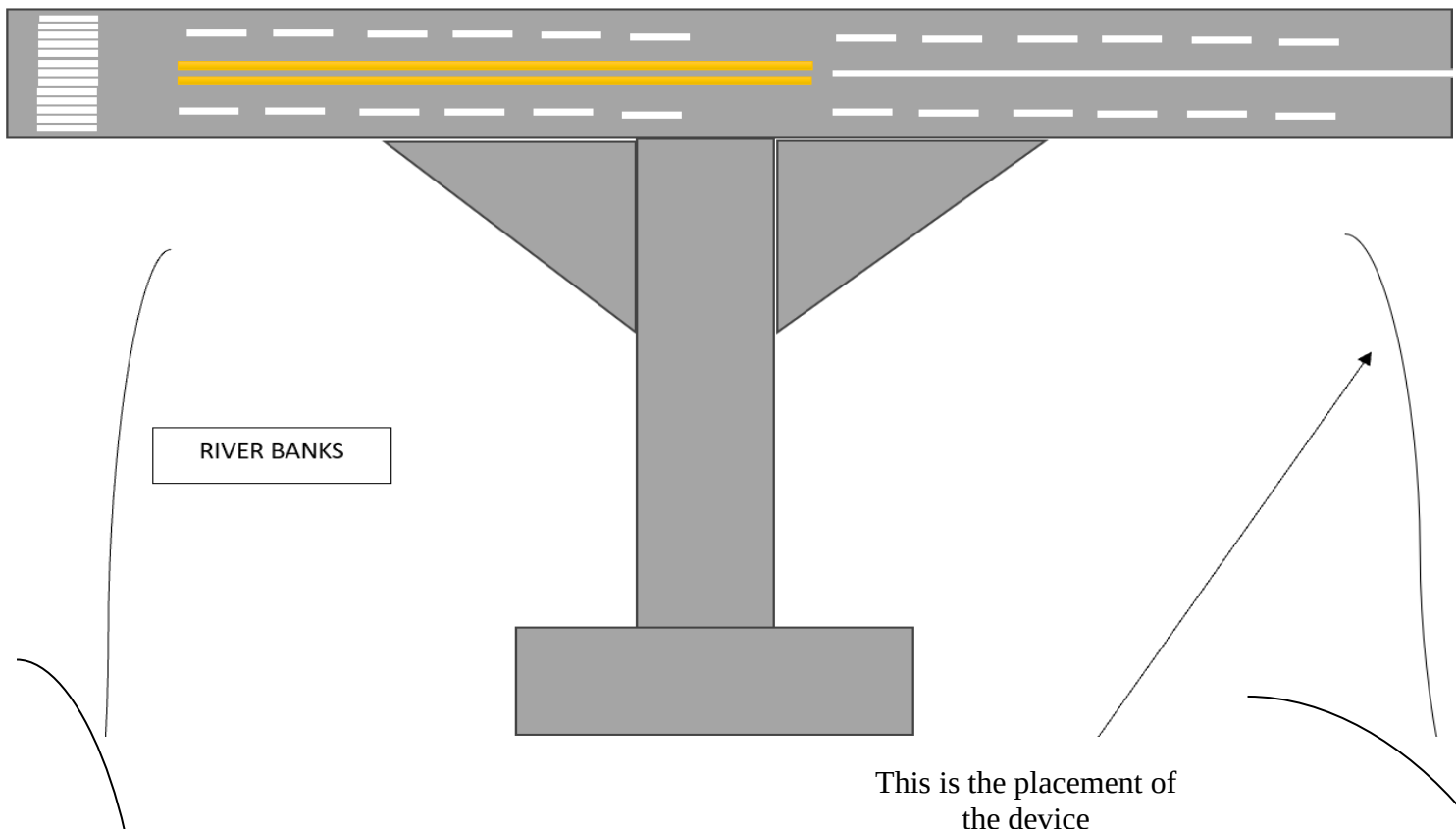
The SafeNest Flood Monitoring System will be deployed near a riverbank to monitor rising and falling water levels. A protective tube will be installed to house the ultrasonic sensor, shielding it from debris and environmental factors while also maintaining an accurate height for distance measurements. The device will be positioned at the river's edge, where fluctuations in water level can be consistently tracked. The sensor will be mounted at a

calibrated height within the tube, enabling accurate data collection. To ensure continuous operation, even in remote locations without direct access to electricity, a solar panel will power the system. The ESP32 microcontroller will transmit the collected water level data via Wi-Fi to Firebase, where it will be displayed in real-time on the monitoring dashboard.

Key Objectives:

- **Location:** The device will be placed near the edge of the river, where fluctuations in water level can be consistently measured.
- **Sensor Placement:** The ultrasonic sensor will be mounted at a calibrated height within the protective tube, allowing it to send accurate distance measurements to the ESP32.
- **Power Source:** The system will be powered by a power bank and solar panel alternative to it, which will ensure continuous operation, even in remote areas without direct access to electricity.
- **Data Transmission:** The ESP32 will transmit the collected water level data to Firebase via Wi-Fi, which will then be displayed in real-time on the dashboard for monitoring.

TUMAGA BRIDGE



CHAPTER 5

DEVELOPMENT AND TESTING

5.1 Development

The development of the SafeNest follows an Agile methodology approach, promoting iterative progress and flexibility in addressing real-time challenges. Initial planning and design phases included analyzing requirements, creating use case diagrams, and defining functional specifications, focusing on the integration of IoT hardware and a web-based monitoring platform. The development used tools such as Arduino IDE for programming the ESP32 microcontroller and Visual Studio Code for the web interface. The ESP32 was configured with ultrasonic sensors for precise water level measurements, while MySQL and Firebase were implemented for user data management and real-time monitoring, respectively. Key stages included hardware integration, software debugging, and usability testing. Challenges, such as sensor calibration and optimizing real-time data transmission, were addressed through rigorous testing and stakeholder feedback. Roles included hardware configuration by the technical team and front-end design by web developers, ensuring seamless interaction between hardware and software components.

5.2 Integration of Technology

The SafeNest project effectively integrated diverse technologies to create a real time flood monitoring system. The ESP32 microcontroller served as the central hub, gathering data from ultrasonic sensors and transmitting it to a Firebase cloud platform for real-time monitoring. The web application utilized HTML, CSS, JavaScript, and PHP for the user interface and MySQL for managing user authentication and historical records. Communication between hardware and the cloud was facilitated through Wi-Fi, while APIs like Twilio enabled SMS alerts for critical flood levels. The system architecture linked IoT devices with a web-based dashboard, ensuring synchronized data flow and user access to vital information. Emphasis was placed on security measures such as input validation and encryption to protect user credentials and data integrity, ensuring reliable and confidential system operation.

5.3 Testing

The primary goal of this testing phase is to assess the system's ability to transmit water level data from the ESP32 microcontroller to Firebase and update the dashboard with minimal delay. This test will evaluate whether the system can meet real-time data transfer and display requirements, which are crucial for effective flood monitoring.

- **Testing Environment**
 - Empty Aquarium simulates as the river

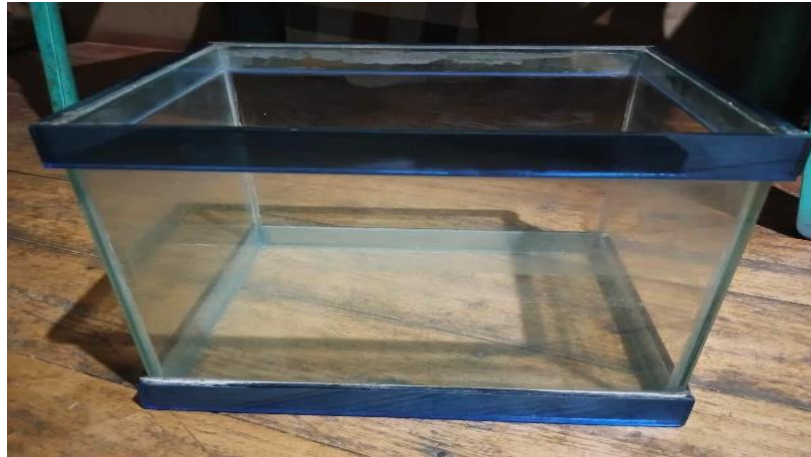


Figure 4.3.12.1 Testing Environment – Empty Aquarium

- SafeNest device with ultrasonic sensor (w/o water)



Figure 4.3.12.2 Testing Environment – SafeNest Device with Ultrasonic Sensor

The testing environment consists of both hardware and software components configured to simulate real-time water level monitoring. The hardware setup includes the ESP32 microcontroller connected to an ultrasonic sensor (HC-SR04), which is responsible for measuring the distance to the water

surface. The calibrated height of the sensor was set to 32 cm where y is the water level that is measured and displayed in the dashboard and R is the raw data that is displayed in the Liquid Crystal Display (LCD). To accurately measure and display the water level on the dashboard, the following formula is used:

$$\text{Water Level (y)} = \text{Sensor Height} - \text{Raw Distance Data (R)}$$

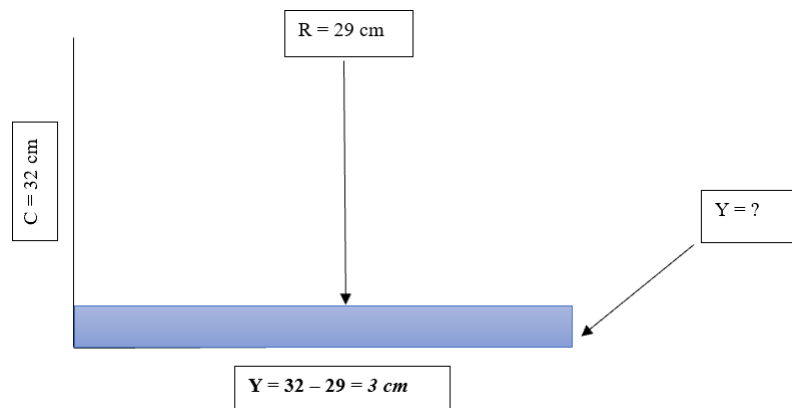


Figure 4.3.12.3 Testing Environment

In this following scenario, y = water level (cm), c = is the height of the sensor from the ground and r = height of water sensor to the surface of the water. The basis of the data is being displayed in the dashboard is the variable y wherein calculated using the formula above.

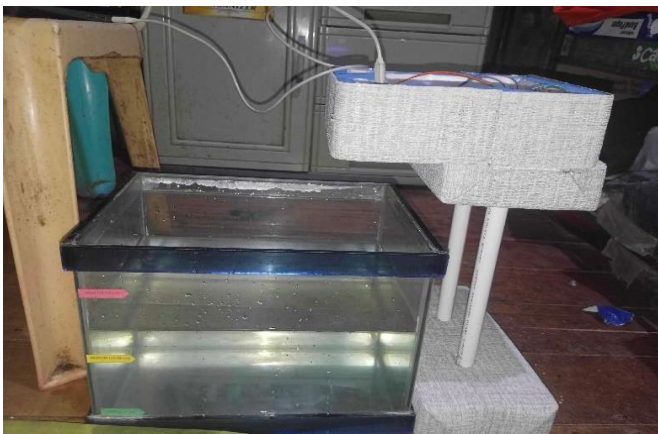


Figure 4.3.12.4 Testing Environment



Figure 4.3.12.5 Testing Environment

In this testing process, several key metrics will be observed to assess the system's performance. First, the Data Transfer Time will be measured, which is the time taken from when the water level changes to when the data is transmitted to Firebase. This ensures that data is quickly captured and sent to the cloud for storage. Next, the Dashboard Update Time will be tracked, which refers to the time it takes for the dashboard to retrieve and display the

updated water level data from Firebase. Finally, the Overall Responsiveness of the system will be evaluated by combining the data transfer and dashboard update times to calculate the total delay between a change in water level and its reflection on the dashboard.

- Metrics to Track

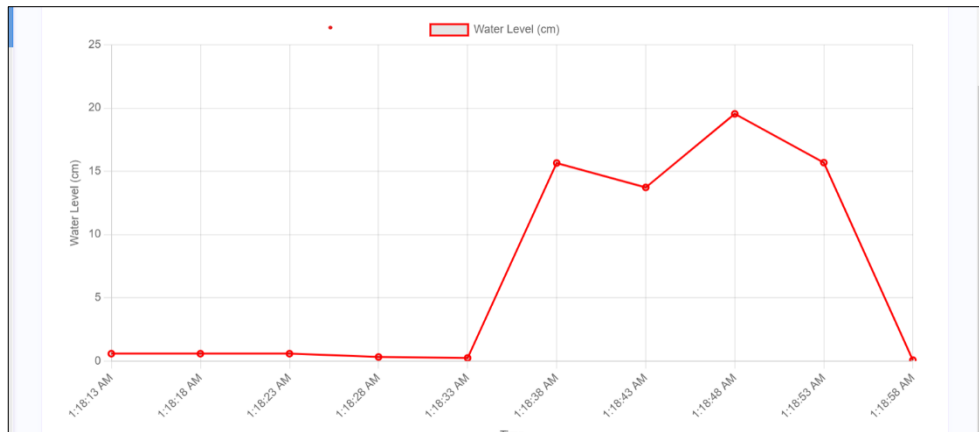


Figure 4.3.12.6 Testing Environment – Metrics to Track

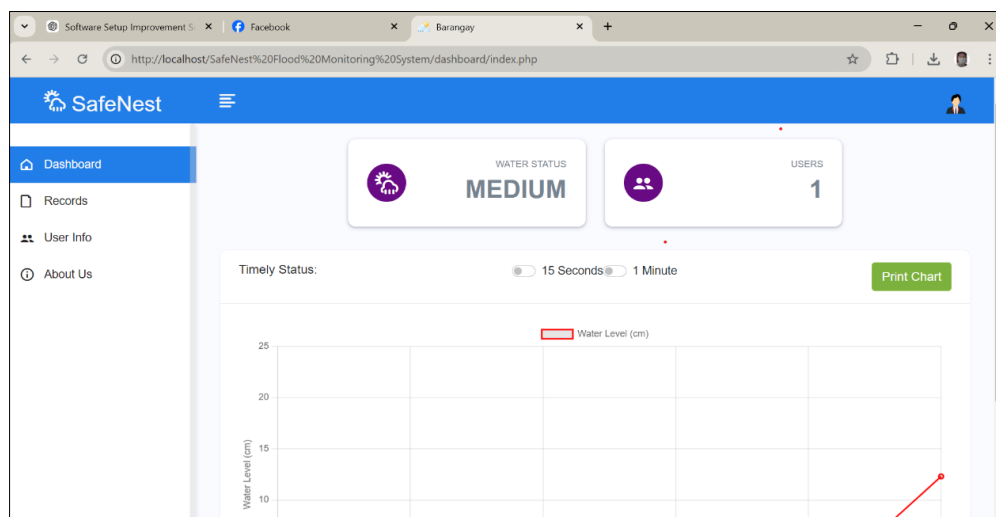


Figure 4.3.12.7 Testing Environment – Metrics to Track

The dashboard, built with Visual Studio Code, retrieves the water level data from Firebase and displays it in real-time for monitoring purposes. The entire system is dependent on a reliable network to minimize any delays during data transmission and dashboard updates. All components are tested under normal indoor conditions to simulate rising and falling water levels in a controlled environment. The system is expected to update water levels in real-time, ideally within 5 seconds or less. If it takes longer, there may be issues like network delays or processing inefficiencies. In such cases, further investigation and optimization would be needed to ensure the system provides timely water level monitoring. Additionally, a toggle switch is use to give

interval to the data that is displayed in the dashboard which makes the data more clear or convenient to visualize.

Test	Condition	Measured Time (seconds)	Success/Failure	Comments/Observations
Data Transfer Time	Normal network conditions	3.0	Success	Data transferred quickly to Firebase.
Data Transfer Time	Simulated network delay	7.0	Failure	High latency observed; network issues noted.
Dashboard Update Time	Normal conditions	5.0	Success	Dashboard updated in real-time.
Dashboard Update Time	High load on the system	6.5	Failure	Delayed update due to system load.
Overall Responsiveness	Normal conditions	4.0	Success	System responds well under typical conditions.
Overall Responsiveness	Network interference simulated	8.0	Failure	Significant delays observed due to interference.

Table 4.3.12 Testing

Testing is a crucial phase in the development process to ensure the system functions correctly and meets user requirements. This section is divided into three sub-sections to cover different testing stages:

5.3.1 Alpha Testing

During Alpha Testing, the team conducted the initial evaluation of the system. This phase enabled them to verify that all components functioned as intended and to identify any technical issues that required attention.

The researchers focused on the following:

- **Hardware Reliability**

- Power Management: Testing the system's ability to operate under different power conditions, including backup and power fluctuations.
 - Connectivity Stability: Evaluating the reliability of Wi-Fi under various environmental conditions.
 - Sensor Calibration: Ensuring the sensors were accurately detecting and consistently under various conditions
- **Software Usability and Functionality**
 - System Latency: Measuring the time, it takes for data to be collected, transmitted, and displayed in real-time.
 - Error Handling: Testing how the system manages incomplete data, sensor disconnections, or unexpected inputs.
 - Data Integrity: Encompasses verifying the correctness, consistency, and reliability of the data collected by the sensors, transmitted from the microcontroller to the cloud, and stored/displayed in the website.
- **Alert Mechanism**
 - Threshold Testing: Testing the water level thresholds to ensure alerts are triggered appropriately.
 - Notification Accuracy: Ensuring notifications are sent to the right recipients with clear and actionable messages.
 - Redundancy Checks: Test fail-safes if primary alert mechanisms fail
- **User Interface (UI) and user Experience (UX)**
 - Ease of Use: Testing the intuitiveness of the web dashboard and the configuration interface, the captive portal.
 - Accessibility: Ensuring the system is accessible on various devices.



Figure 5.3.1.1 Alpha Testing



Figure 5.3.1.2 Alpha Testing

5.3.2 Beta Testing

Beta testing was conducted to evaluate the effectiveness and usability of the "SafeNest: Contactless Based Real-Time Flood Detection and Monitoring System" in a real-world environment. Barangay officials participated in this phase to test the system, ensuring that all functionalities met the intended objectives. The testing process is visually documented in Figure 5.3.2.1 to Figure 5.3.2.4.

- The primary objectives of beta testing were:
 - To validate the accuracy and reliability of water level detection.
 - To ensure real-time updates were displayed on the dashboard.
 - To test the functionality of SMS alerts for notifying stakeholders of potential floods.
 - To evaluate user-friendliness from the perspective of barangay officials.
- Procedure
 - System Interaction: Barangay officials interacted with the system by monitoring water levels displayed on the dashboard.
 - Simulated Scenarios: Controlled water levels were used to trigger thresholds for "Low," "Medium," and "High" statuses, activating the LED indicators and SMS alerts.
 - Dashboard Usage: Officials accessed the dashboard to generate reports and review historical flood data.
 - Feedback Collection: Evaluation forms were distributed to gather user impressions on system performance, usability, and areas for improvement.



Figure 5.3.2.1 Beta Testing



Figure 5.3.2.2 Beta Testing



Figure 5.3.2.3 Beta Testing



Figure 5.3.2.4 Beta Testing

- Figures
 - Figure 5.3.2.1: Barangay officials examining the SafeNest hardware setup.
 - Figure 5.3.2.2: Dashboard in use, showing live water level monitoring.
 - Figure 5.3.2.3: Officials testing the SMS alert system.
 - Figure 5.3.2.4: Feedback and evaluation session with the barangay officials.

SAFENEST: CONTACTLESS BASED REAL-TIME FLOOD DETECTION AND MONITORING SYSTEM				
Questions	Strongly Agree	Agree	Disagree	Strongly Disagree
The system provides accurate real-time flood data.	✓			
The flood monitoring dashboard is easy to navigate.	✓			
The visual graphs for water levels are clear and informative.	✓			
The SMS alert system provides timely notifications of flooding risks.	✓			
The ultrasonic sensor used in the system works reliably.	✓			
The system's user interface is visually appealing.	✓			
The flood monitoring system is easy to use for all intended users.	✓			
The training provided for using the system is sufficient.		✓		
The system significantly improves disaster preparedness in the community.	✓			
I am satisfied with the overall performance of the flood monitoring system.	✓			
The notifications sent are easy to understand.	✓			
The system enhances the safety of residents in flood-prone areas.	✓			
The water level thresholds are appropriately calibrated for timely alerts.		✓		
The system is responsive even under high network load.		✓		
The real-time data updates meet user expectations.	✓			

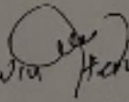
 **Darwin** *Hernandez* support staff
 Brig. Tumaga

Figure 5.3.2.5 Evaluation form

Pa/ M Magalona

SAFENEST: CONTACTLESS BASED REAL-TIME FLOOD DETECTION AND MONITORING SYSTEM				
Questions	Strongly Agree	Agree	Disagree	Strongly Disagree
The system provides accurate real-time flood data.	/			
The flood monitoring dashboard is easy to navigate.		/		
The visual graphs for water levels are clear and informative.	/			
The SMS alert system provides timely notifications of flooding risks.		/		
The ultrasonic sensor used in the system works reliably.	/			
The system's user interface is visually appealing.		/		
The flood monitoring system is easy to use for all intended users.	/			
The training provided for using the system is sufficient.	/			
The system significantly improves disaster preparedness in the community.	/			
I am satisfied with the overall performance of the flood monitoring system.	/			
The notifications sent are easy to understand.	/			
The system enhances the safety of residents in flood-prone areas.	/			
The water level thresholds are appropriately calibrated for timely alerts.	/			
The system is responsive even under high network load.	/			
The real-time data updates meet user expectations.	/			

Figure 5.3.2.6 Evaluation form

Banyuwir

Mila M. PINKIAN - YVANDETT Tamaya 2.C

SAFENEST: CONTACTLESS BASED REAL-TIME FLOOD DETECTION AND MONITORING SYSTEM				
Questions	Strongly Agree	Agree	Disagree	Strongly Disagree
The system provides accurate real-time flood data.	/			
The flood monitoring dashboard is easy to navigate.	/			
The visual graphs for water levels are clear and informative.	/			
The SMS alert system provides timely notifications of flooding risks.	/			
The ultrasonic sensor used in the system works reliably.	/			
The system's user interface is visually appealing.		/		
The flood monitoring system is easy to use for all intended users.	/			
The training provided for using the system is sufficient.	/			
The system significantly improves disaster preparedness in the community.	/			
I am satisfied with the overall performance of the flood monitoring system.		/		
The notifications sent are easy to understand.	/			
The system enhances the safety of residents in flood-prone areas.	/			
The water level thresholds are appropriately calibrated for timely alerts.	/			
The system is responsive even under high network load.	/			
The real-time data updates meet user expectations.	/			

Figure 5.3.2.7 Evaluation form

5.3.3 User Acceptance Testing

Barangay officials expressed satisfaction with the system's functionality and user experience. Key areas noted in the evaluation form feedbacks included:

- High accuracy and reliability of water level detection.
- Timely and efficient real-time updates on the dashboard.
- Clear and actionable SMS alert notifications.
- User-friendly and intuitive dashboard interface.
- Effective visual indicators for quick water level assessment.
- Practicality for integration with disaster management workflows.
- Readiness for deployment in flood-prone areas.

5.4 Implementation Plan

The implementation plan outlines the steps needed to deploy the SafeNest: Contactless Based Real-time Flood Detection and Monitoring System in a real-world environment. This section is divided into two sub-sections to cover different aspects of implementation:

5.4.1 Infrastructure/Deployment

The deployment of the SafeNest system requires a robust infrastructure to ensure reliable and efficient operation. The server environment involves a cloud-based Firebase service for real-time data storage and MySQL for user management, hosted on a Node.js-based web server. The server specifications include at least 8GB of RAM, a quad-core processor, and 500GB of storage to handle the system's data processing and storage needs. A stable internet connection with minimum speeds of 10 Mbps is essential for real-time data transmission from the ESP32 microcontroller to the cloud. The deployment process begins with configuring the server environment, including installing necessary software and setting up the Firebase and MySQL databases. The application is then deployed using secure protocols, followed by configuring the ESP32 devices to transmit data. Backup and recovery are managed using automated daily backups of critical data stored in Firebase and MySQL, with recovery procedures in place for system restoration. High availability is ensured through redundant servers and load balancing, while scalability is supported by the cloud infrastructure to accommodate increased demand. The deployment timeline spans five days, allowing for server setup, hardware integration, testing, and system validation, with minimal downtime during the transition to operational status.

5.4.2 Processes/Policies/Personnel

Effective implementation and management of SafeNest require clear processes and policies, supported by a dedicated team. Deployment and maintenance are handled by IT staff responsible for configuring the server, integrating IoT hardware, and monitoring system performance. Academic advisors and administrators oversee the

system's alignment with community goals, ensuring it meets safety and accessibility requirements. Training programs are provided to users, including barangay officials, to familiarize them with system functionalities, such as real-time monitoring and alert generation. Policies focus on data security, enforcing strong user authentication, role-based access control, and encrypted data transmission. Privacy measures ensure that sensitive information, such as user credentials and water level data, is protected. Regular system monitoring and scheduled maintenance are critical to maintaining reliability, with IT personnel addressing issues promptly and applying updates to improve performance. These processes ensure the system remains operational, scalable, and aligned with the needs of Barangay Tumaga's emergency response framework.

CHAPTER 6

RESULT AND DISCUSSIONS

6.1 Introduction

This chapter presents the results of the development, testing, and evaluation of the "*SafeNest: Contactless Based Real-Time Flood Detection and Monitoring System.*" It assesses the effectiveness of the system in meeting its objectives through alpha, beta, and user acceptance testing. Additionally, the aim is to determine whether the objectives of real-time flood monitoring, data transmission, and user accessibility were successfully achieved.

6.2 Summary of Findings

The SafeNest system met its core objectives, integrating IoT technology and cloud-based system effectively. Testing validated the functionality and performance of hardware and software components.

Hardware Integration

- ESP32 microcontroller accurately processed data from the ultrasonic sensor.
- LED indicators effectively displayed water levels (low, medium, high).
- The LCD1602 provided real-time feedback with minimal delay.

Software Performance

- Firebase handled real-time data storage and retrieval efficiently.
- The dashboard displayed water levels with an update delay of under five seconds in normal conditions.
- Twilio-powered SMS notifications were timely and accurate

Testing Results

- **Alpha Testing:** Validated hardware reliability and software functionality under controlled conditions.

- **Beta Testing**

The beta testing phase validated the "SafeNest: Contactless Based Real-Time Flood Detection and Monitoring System" as a dependable and user-friendly tool for flood monitoring. An IT staff member supervised the testing process, while barangay officials, who were all non-technical users, assessed the system's functionality and usability. The system demonstrated high accuracy in water level detection, efficient real-time updates, and reliable SMS alert functionality. The dashboard interface was praised for its simplicity, enabling easy navigation and quick access to critical data. Visual indicators, such as LED lights, along with features for generating reports and analyzing historical data, were highly appreciated by the officials. All participants expressed their readiness to adopt the system for flood-prone areas, highlighting its potential to significantly enhance disaster preparedness and community safety. The system successfully met its objectives, with minor suggestions for improved data visualization and additional training for non-technical users.

- **User Acceptance Testing:** Highlighted a satisfaction with system usability and practical application.

Challenges

- Minor delays were noted under high system load or network interference.
- Initial sensor calibration required iterative adjustments for accuracy.

Testing Result:

Metrics: Metrics like data transfer time (3) three seconds under normal conditions and dashboard update time (5) five seconds under load confirmed the system's performance.

Observations:

- Delay under high load or network interference were noted.
- Sensor calibration was effective, with consistent accuracy.

Evaluation Result:

Graphical User Interface (GUI) and User Experience (UX): The system's graphical interface and overall user experience were evaluated by barangay officials and project stakeholders. Feedback highlighted the following:

- **Clarity:** The dashboard layout effectively presented water level data in graphical form, categorized by thresholds (low, medium, high). Clear color indicators (green, yellow, red) enhanced visibility.
- **Ease of Navigation:** User roles (e.g., Admin, Kagawad, SK Kagawad) were assigned appropriate permissions, simplifying interactions based on responsibility.
- **Aesthetic Appeal:** The clean and modern design, coupled with responsive web elements, ensured accessibility across various devices, including smartphones and tablets.
- **Suggestions for Improvement:** Participants suggested adding a dark mode for better usability in low-light conditions and customizable thresholds for different barangay requirements.

System Usability: The system's usability was tested against predefined criteria, including accessibility, learnability, and efficiency. Key findings:

- **Accessibility:** The system was accessible through common browsers and required minimal technical expertise for operation. Officials were able to log in and interact with the dashboard seamlessly.
- **Learnability:** Participants reported ease in understanding the system's functionalities after a short orientation session, with most users confident in operating the system independently.

- **Efficiency:** Time taken to access real-time data and perform actions, such as generating reports or sending alerts, was within acceptable limits.

Quantitative Usability Metrics (Based on User Feedback)

Usability Aspect	Average Score (Out of 5)
Ease of use	4.3
Navigation Clarity	4.3
Information Readability	5.0
Response Speed	4.3

System Functionalities: Each core functionality of the system was rigorously tested to ensure alignment with project objectives. Results are summarized below:

Functionality	Status	Observations
Real-time Monitoring	Passed	Accurate water level detection and display on the dashboard.
SMS Notification Alerts	Passed	Notifications were timely and accurately sent to recipients.
Role-based Access Control	Passed	User roles and permissions were enforced effectively.
Data Visualization	Passed	Historical data and graphical trends were clearly presented.
Report Generation	Passed	Reports could be generated and exported in multiple formats (PDF, CSV).
Database Reliability	Passed	No data loss or corruption observed during testing.

Summary of Feedback from Users

- **Strengths:** High reliability in real-time monitoring, intuitive interface, and the practicality of SMS alerts.

- **Areas for Improvement:** Enhanced offline capabilities for areas with intermittent internet connectivity and greater sensor range for wider coverage.

CHAPTER 7

CONCLUSION AND RECOMMENDATIONS

7.1 Conclusions

The SafeNest: Contactless Based Real-Time Flood Detection and Monitoring System successfully addressed the key objectives of providing accurate, timely, and accessible flood monitoring for Barangay Tumaga. Through the integration of IoT technology, cloud storage, and user-friendly interfaces, the system demonstrated its capability to aid disaster risk management efforts. The following conclusions are drawn from the project:

- **System Performance:** The system reliably measured water levels using ultrasonic sensors and displayed data on a real-time dashboard with minimal delay (3-5 seconds under normal conditions).
- **Usability:** Feedback from barangay officials and stakeholders indicated a high level of satisfaction with the system's design, usability, and efficiency.
- **Alert Mechanism:** SMS notifications powered by Twilio proved effective for disseminating critical flood warnings.
- **Impact:** The project has the potential to improve community resilience by enabling proactive disaster management and timely decision-making by local authorities.

This system can serve as a blueprint for similar initiatives in other flood-prone areas, showcasing the scalability and adaptability of IoT-based flood monitoring systems.

7.2 Recommendations

To further enhance the system and its impact, the following recommendations are proposed:

- **System Improvements**
 - **Hardware:** Improve the durability of hardware enclosures to withstand extreme weather conditions. Consider integrating solar panels to ensure uninterrupted operation in remote areas.

- **Software:** Optimize the system for handling high data loads to minimize delays during peak usage.
- **Additional Features**
 - **Expanded Sensor Integration:** Incorporate additional sensors (e.g., rainfall, wind speed) to provide comprehensive environmental monitoring.
- **Scaling and Deployment**
 - **Broader Deployment:** Extend the system to other barangays or municipalities prone to flooding.
- **Future Researchers**
 - Investigate alternative technologies for data transmission, such as LoRaWAN, to enhance connectivity in areas with weak internet access.
 - Explore machine learning algorithms to predict flood events based on historical and real-time data.

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